
Denoising the Market

Conditional Score-Based Financial Time-Series Generation with EDM-CSDI

Master's Thesis Defence

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DSBA, Bocconi University

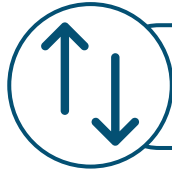
July 20th, 2026

Why?

Complex



Returns are not Gaussian nor i.i.d.



r_t are uncorrelated, yet $|r_t|$ are



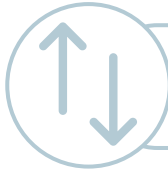
Regime dependent SNR

Why?

Complex



Returns are not Gaussian nor i.i.d.



r_t are uncorrelated, yet $|r_t|$ are



Regime dependent SNR

Useful



Data is scarce and not repeatable



Backtesting and scenario analysis

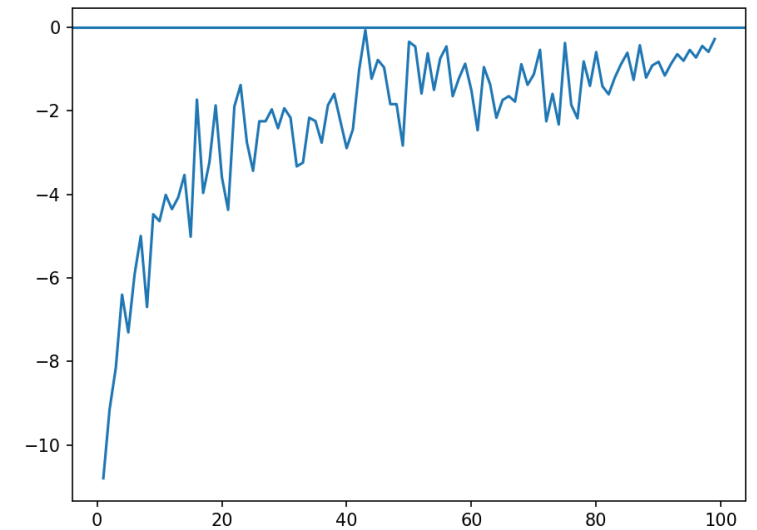
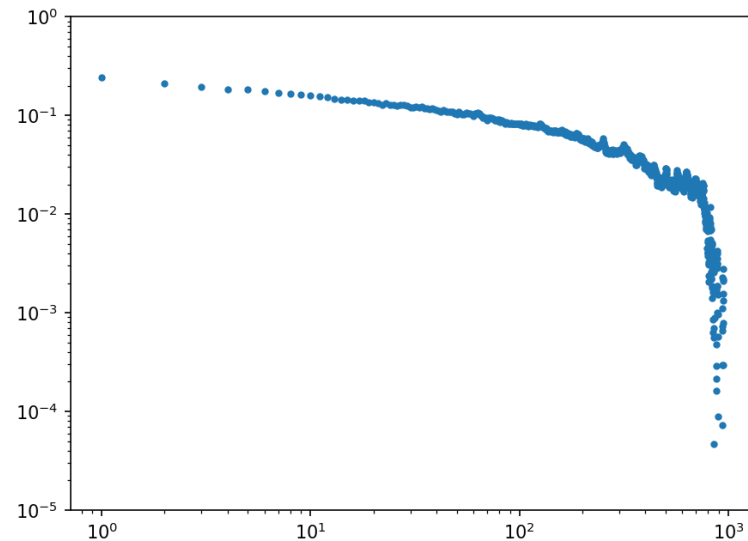
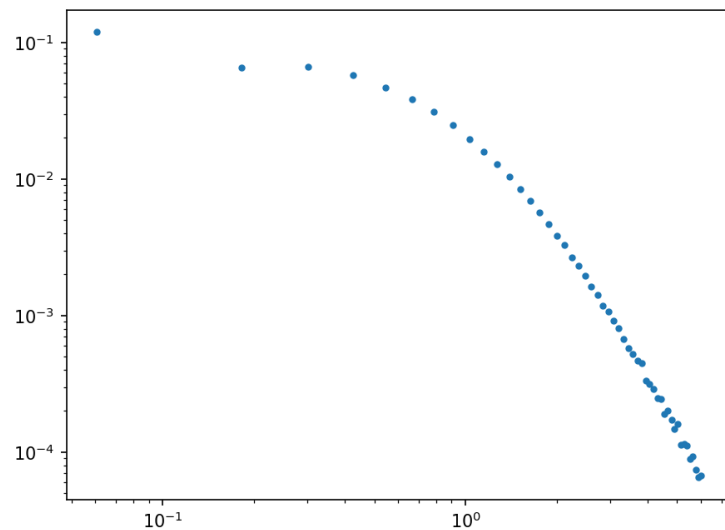


Risk and portfolio management

The empirical gap: what a generator must reproduce

Stylized facts of log-returns $r_t = \log(C_t/C_{t-1})$:

- **Heavy tails:** $P(|r| > x) \sim x^{-\alpha}$, $\alpha \in [3,5]$
- **Volatility Clustering:** $\text{Corr}(|r_t|, |r_{t+k}|) \sim k^{-\beta}$
- **Leverage effect:** $L(k) = E[r_t r_{t+k}^2] / \mathbb{E}[r_t^2]^2 < 0$



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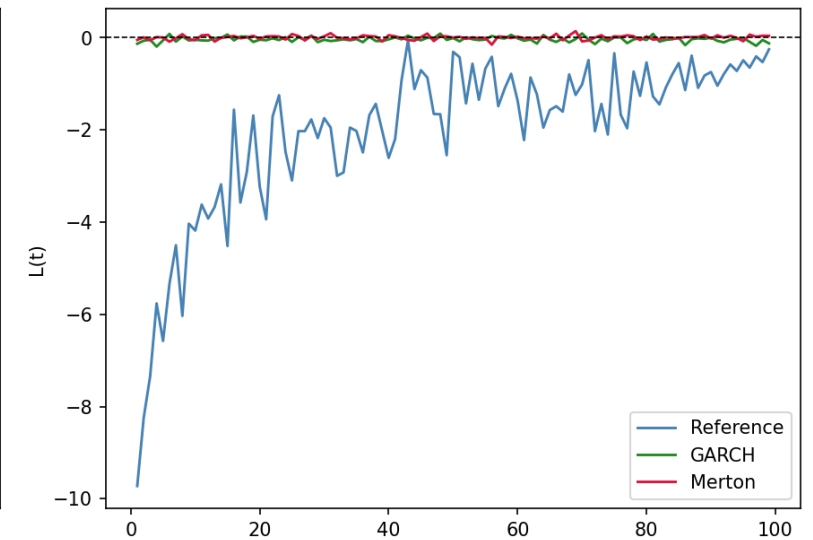
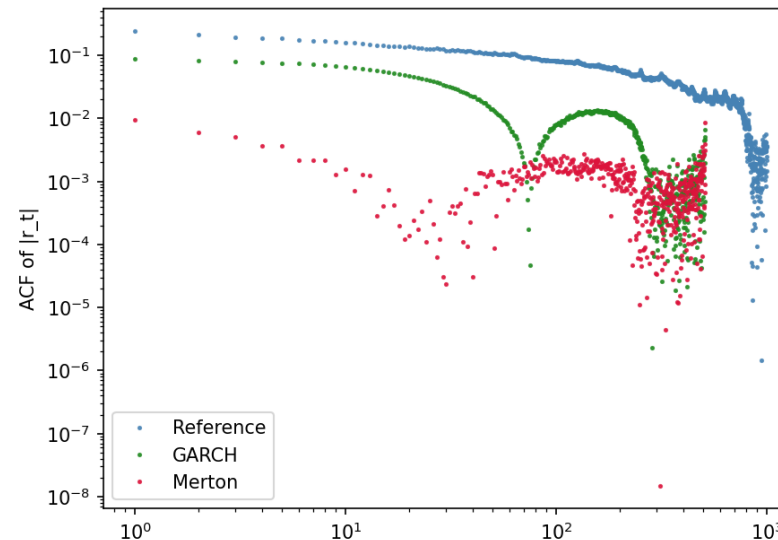
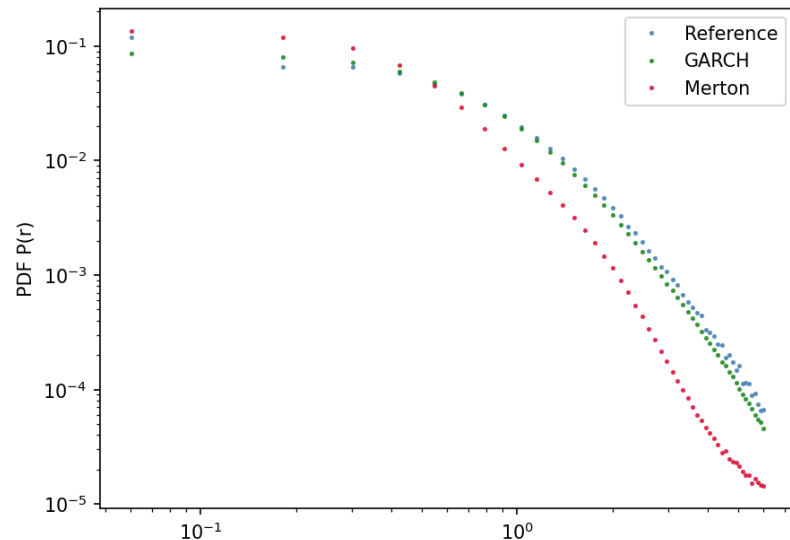
Classical models

GARCH-X

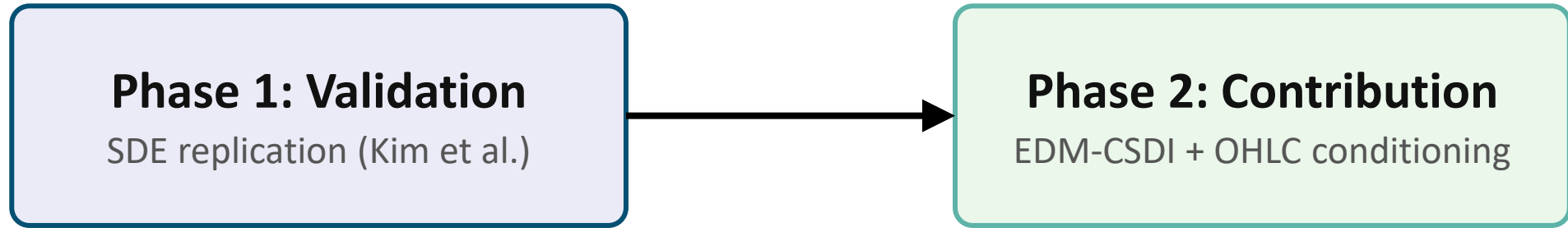
✓ clustering — ✗ leverage effect

Merton-X

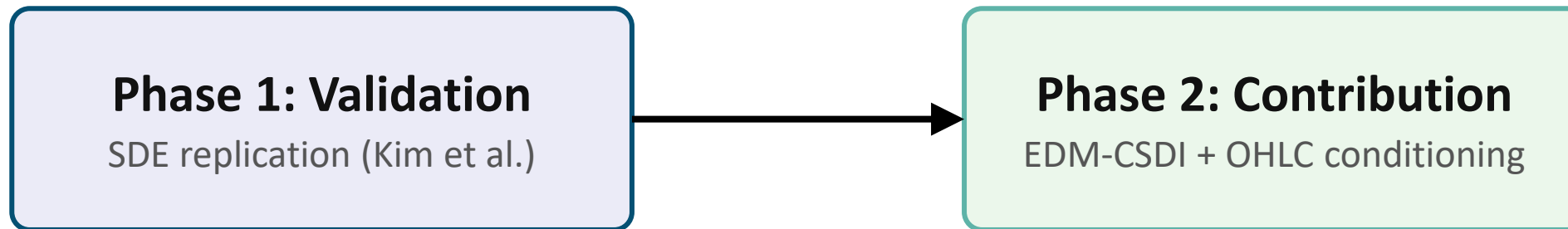
✓ jumps — ✗ volatility clustering



Two research questions, two phases



Two research questions, two phases



RQ1 — Phase 1: Replication

Can a CSDI-based architecture, trained under the SDE families of Kim et al. (VE, VP, GBM), learn the distributional properties of S&P 500 log-returns and reproduce their stylized facts?

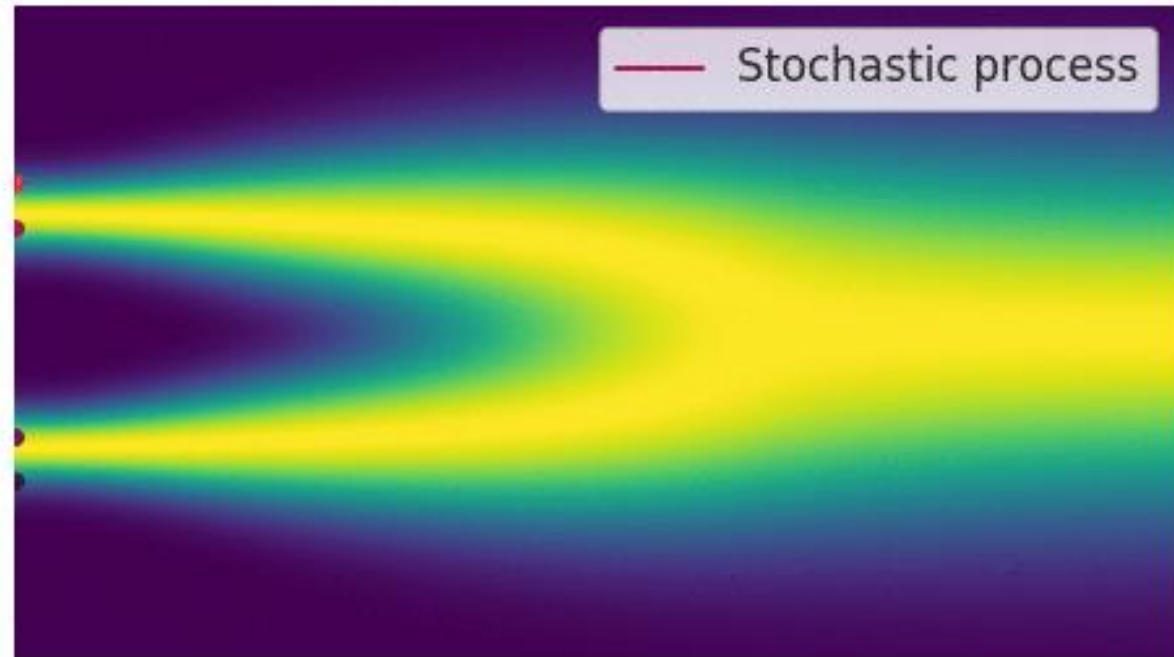
RQ2 — Phase 2: EDM-CSDI

Does the EDM framework of Karras et al., combined with same-day $\{O, H, L\}$ conditioning, improve distributional fidelity and conditional predictive quality?

Diffusion: noising to learn how to denoise

Forward

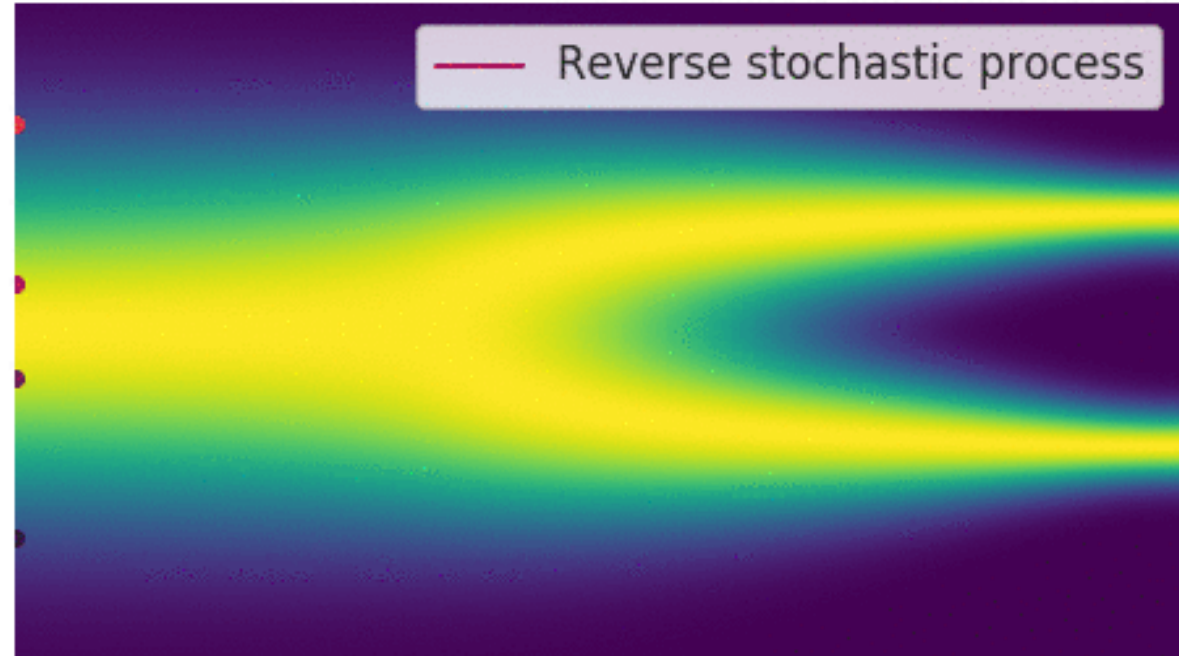
$$dx = f(x, t)dt + g(t)dw$$



Diffusion: noising to learn how to denoise

Reverse

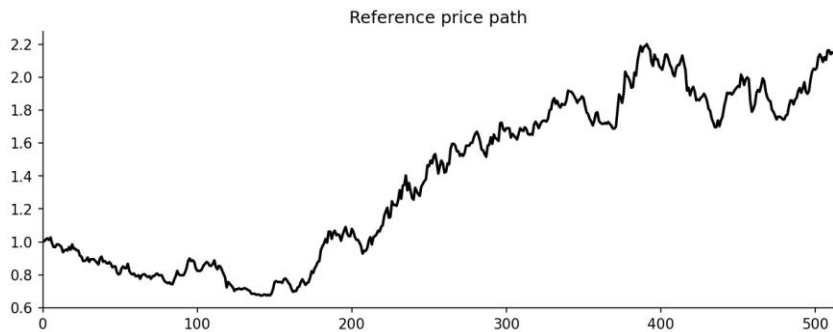
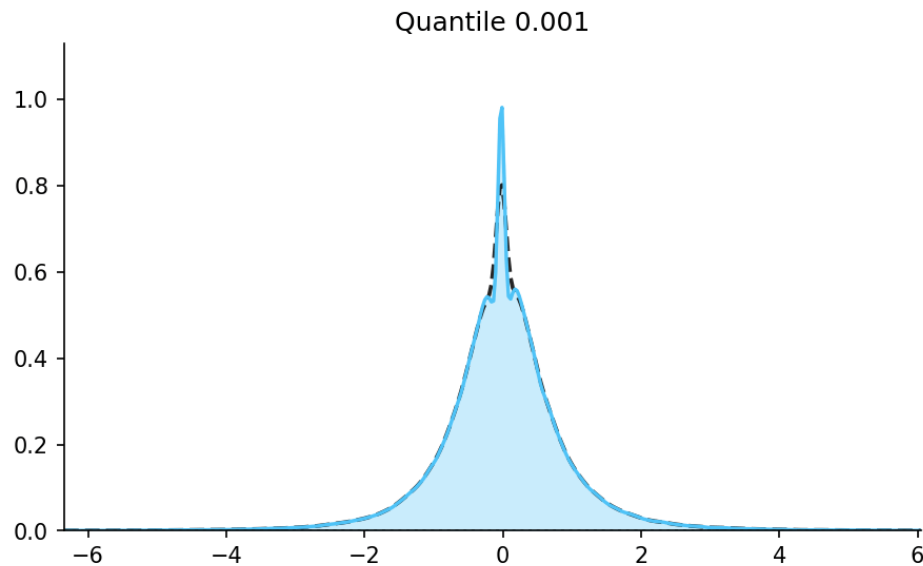
$$dx = [f(x, t) - g(t)^2 \nabla_x \log p_t(x)]dt + g(t)d\bar{w}$$



Diffusion: noising to learn how to denoise

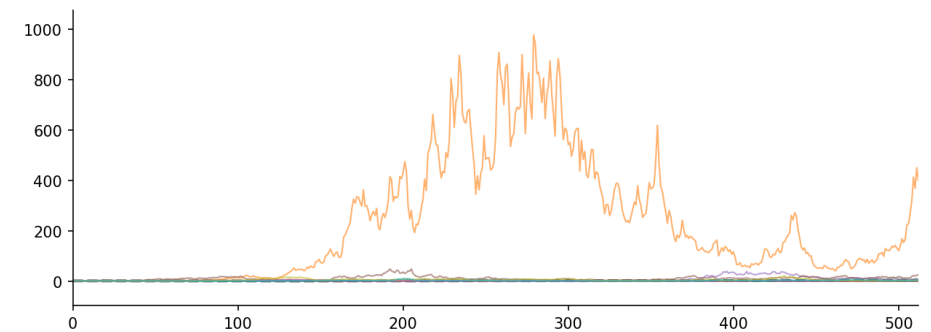
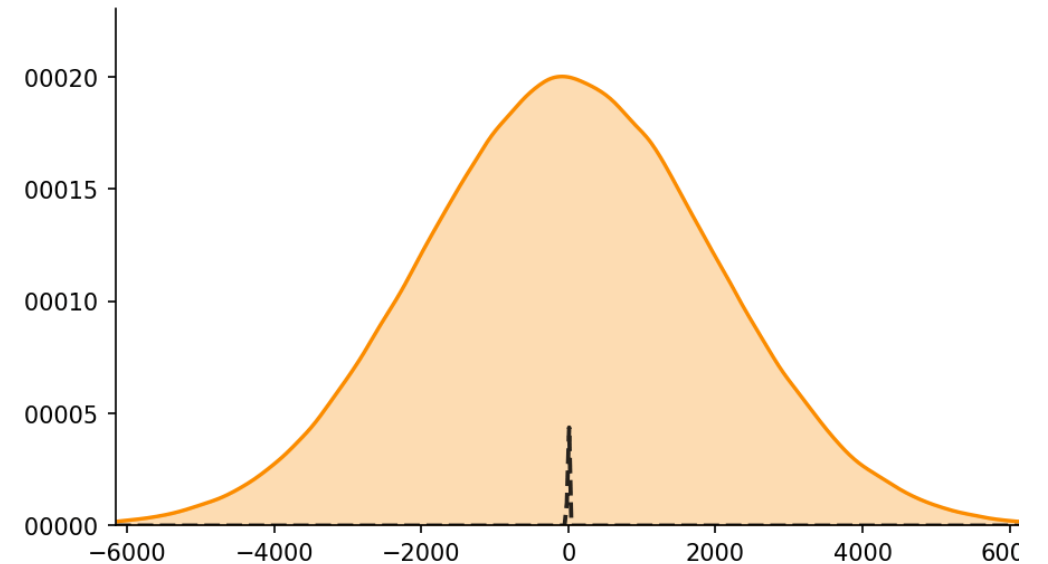
Forward

$$dx = f(x, t)dt + g(t)dw$$



Reverse

$$dx = [f(x, t) - g(t)^2 s_{\theta}(\cdot)]dt + g(t)d\bar{w}$$



Conditioning on same-day Open, High, Low

Phase 1

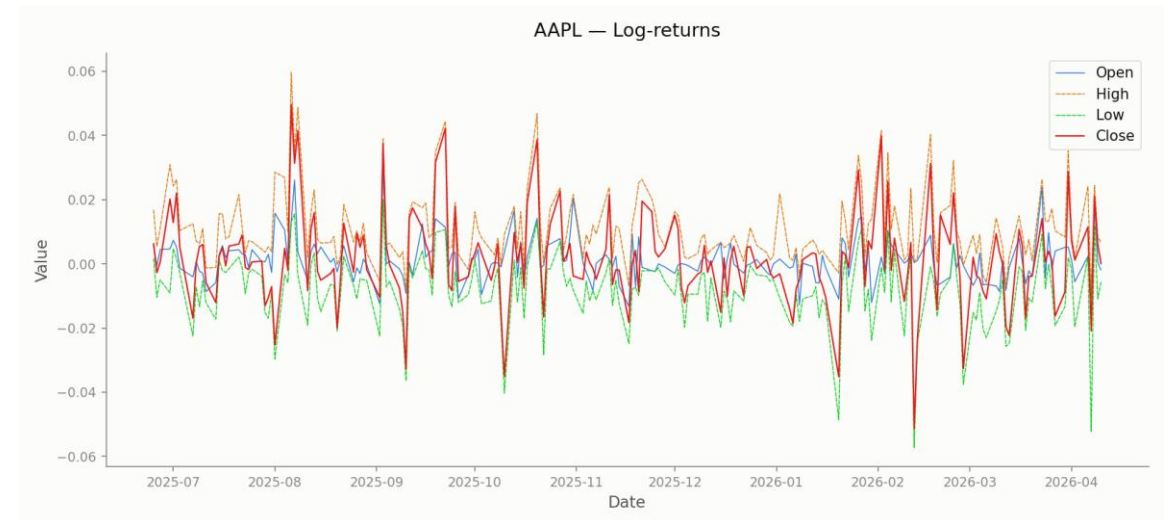
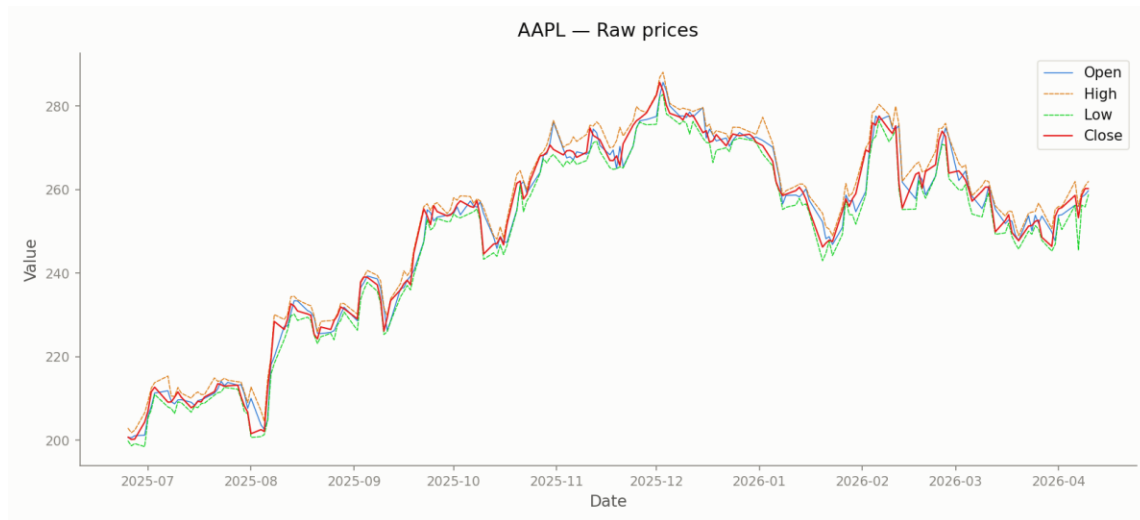
$$\mathcal{C} = \emptyset, \mathcal{T} = \{C\}$$

- **C** : closing price: $\log(C_t/C_{t-1})$

Phase 2

$$\mathcal{C} = \{O, H, L\}, \mathcal{T} = \{C\}$$

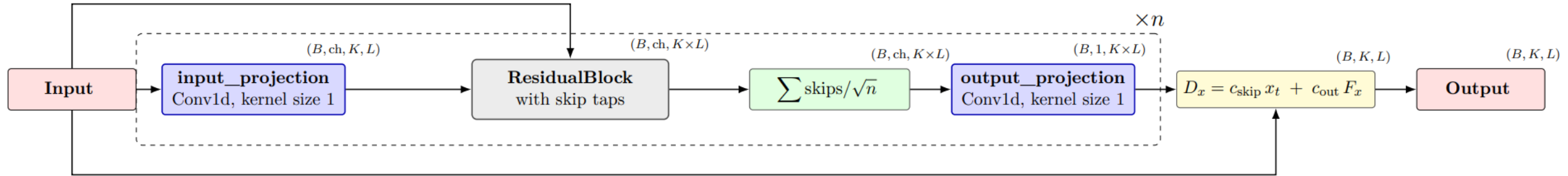
- **C** (closing price): $\log(C_t/C_{t-1})$
- **O** (opening price): $\log(O_t/C_{t-1})$
- **H, L** (intraday high/low): $\log(H_t/C_{t-1})$ / $\log(L_t/C_{t-1})$



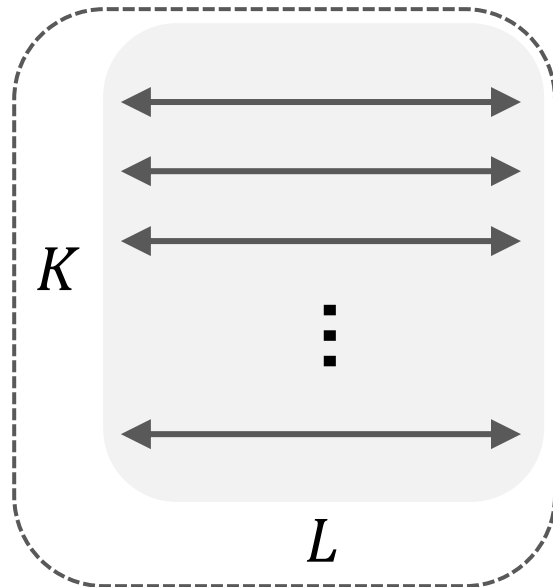
From SDE replication to EDM: summary

	Phase 1 — Replication	Phase 2 — EDM-CSDI
Task	unconditional*, $K = 1$ (close)	predict_close: $\{O, H, L\} \rightarrow \{C\}$, $K = 4$
Objective	$\hat{\epsilon}$ -MSE, $\lambda(\sigma) = \sigma^2$	x_0 -MSE, $\lambda(\sigma) = c_{\text{out}}(\sigma)^{-2}$
Noise	$t \sim \mathcal{U}[0, 1]$, $\sigma(t)$ with ad-hoc $\sigma_{\min}, \sigma_{\max}$	$\ln \sigma \sim \mathcal{N}(P_{\text{mean}}, P_{\text{std}}^2)$, $\sigma_{\text{data}} = 1$
Sampler	1st order	2nd-order Heun

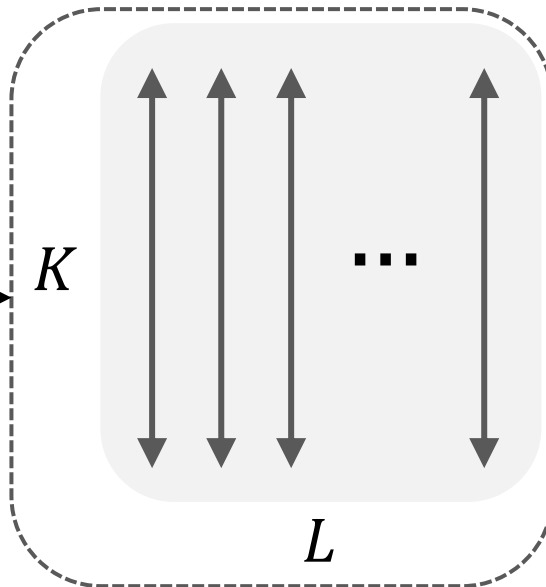
Architecture: CSDI transformer backbone



Temporal Transformer



Feature Transformer



Embeddings

Diffusion step: encodes noise level

Time: encodes calendar position

Feature: encodes channel identity

Positional: encodes sequence order

Data

Data

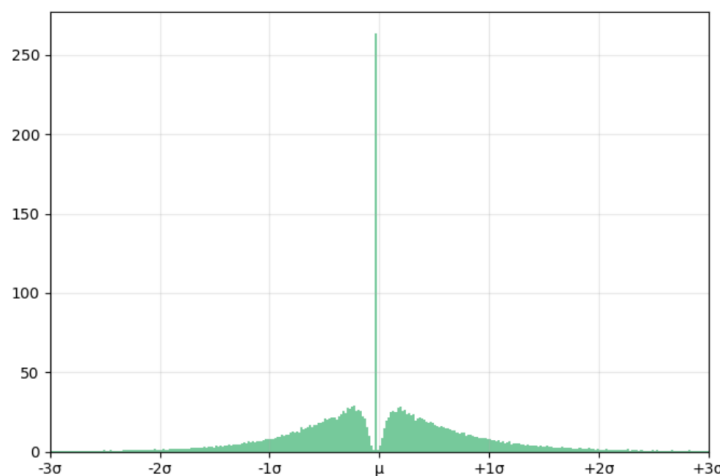
- 210 tickers of S&P500
- ≥ 40 years of trading history
- 7.1928% have zero log-return

Window

- Length:
 - Phase 1: 2048
 - Phase 2: 512
- Stride:
 - Phase 1: 400
 - Phase 2: 100

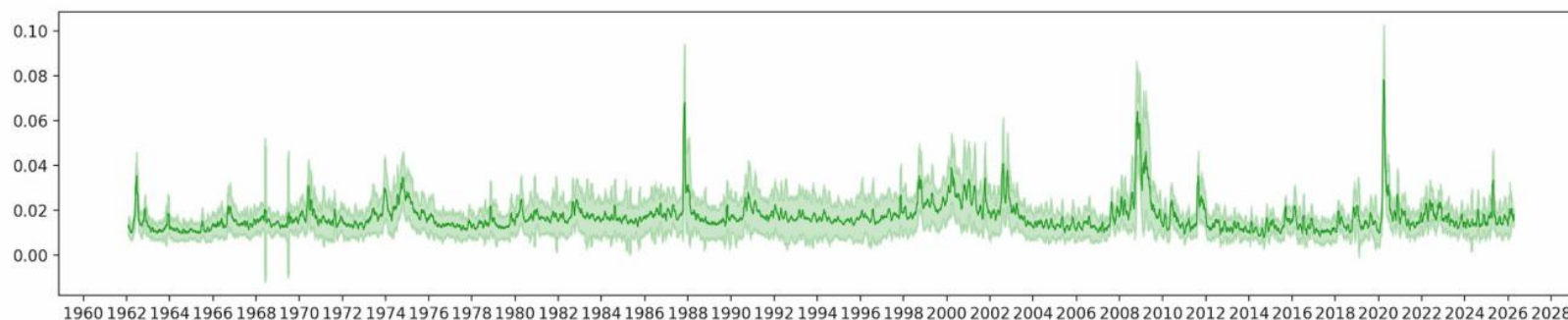
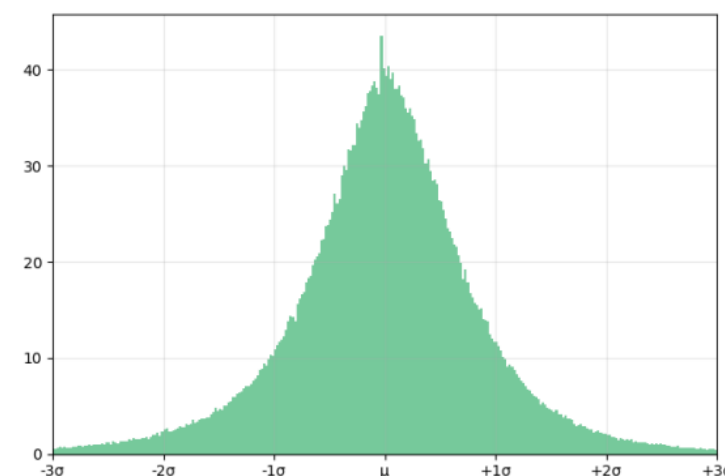
1970 – 1980

e.g. \$ $15 \frac{1}{16}$



2010 – 2020

e.g. \$ 15,0625



Evaluation methodology

Distributional

- **Kolmogorov-Smirnov, D_{KS}**

$$\sup_x |F(x) - G(x)|$$

- **Wasserstein Distance, $W_1(F, G)$**

$$\int_{-\infty}^{\infty} |F(x) - G(x)| dx$$

Conditional

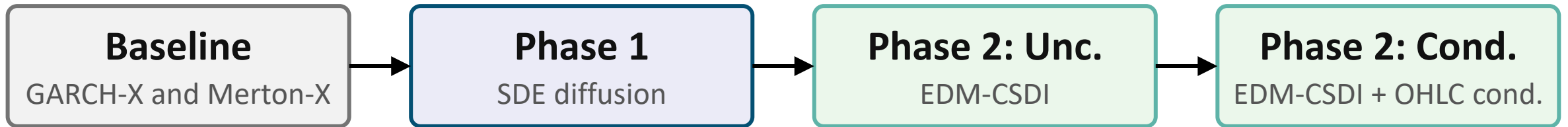
- **Continuous Rank Probability Score, $CRPS(\hat{F}, y)$**

$$\mathbb{E}_{\hat{F}}[|X - y|] - 1/2 \mathbb{E}_{\hat{F}}[|X - X'|]$$

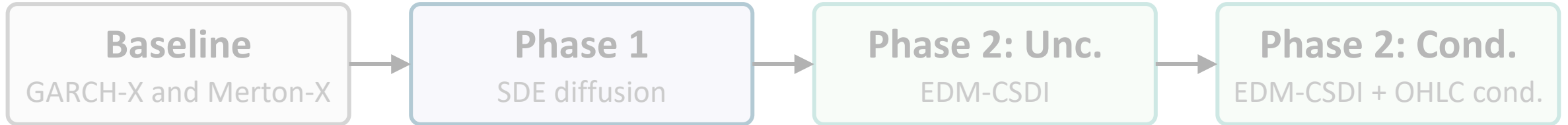
- **Interval Coverage, $cov_{1-\alpha}$**

$$1/N \sum_{t=1}^L \sum_{n=1}^N \mathbf{1}[y_{n,t} \in [q_{\alpha/2}(n,t), q_{1-\alpha/2}(n,t)]]$$

Results: Overview



Results: Overview



Phase 1: SDE Diffusion



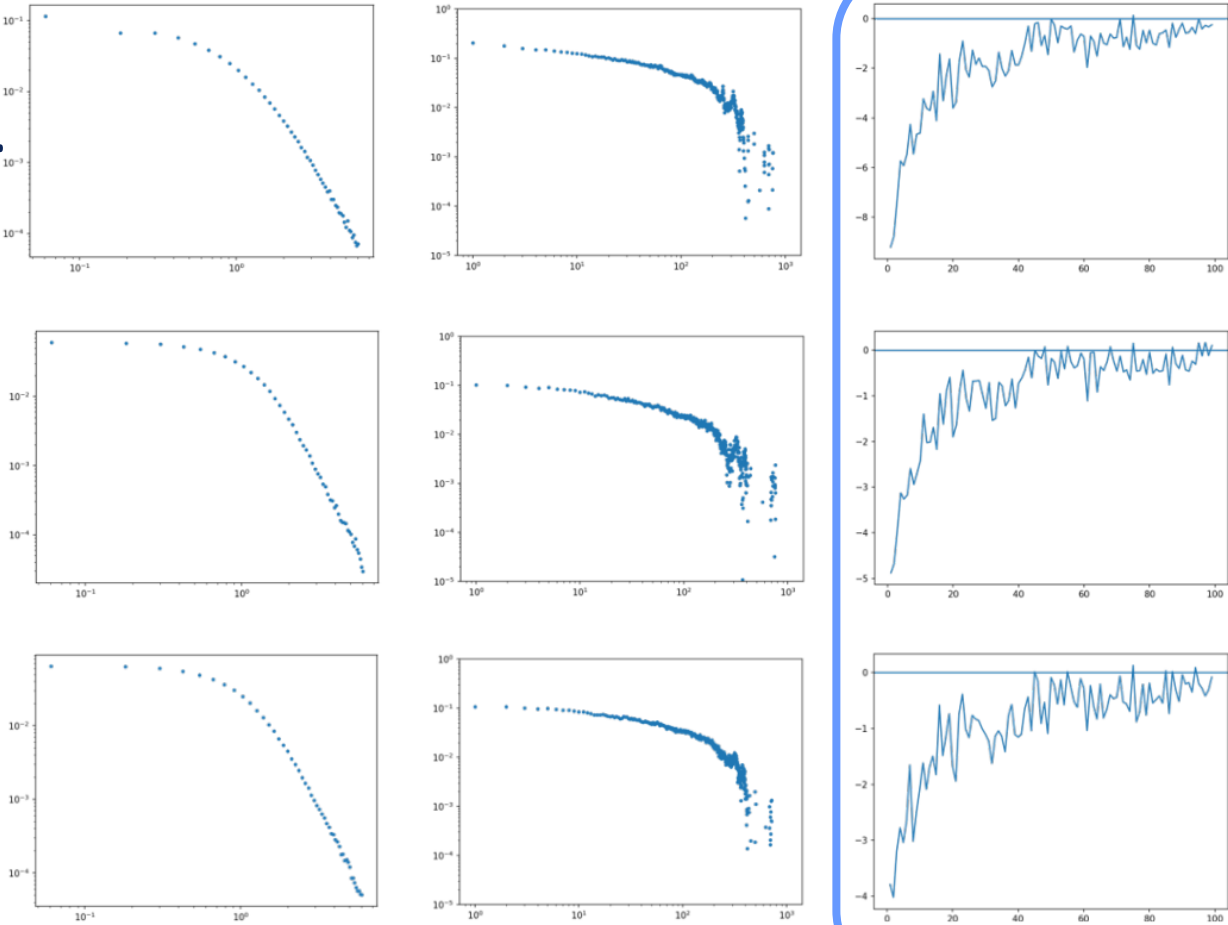
	Reference	VE + Exp.	VP + Cos.
Mean	0.0005	0.0005	0.0003
Std	0.0208	0.0190	0.0235
Skew	-0.5387	-0.0860	-0.2985
Kur	36.0	14.3	15.7
W_1	—	0.002	0.003
D_{KS}	—	0.037	0.060
$\hat{\alpha}$	4.378	4.461	4.368

Table 6.2

Ref.

VE

VP



- Leverage effect is learned partially
- Comparable Tail Exponent
- Poor Skewness recovery
- Kurtosis gap (~ 60%)

Phase 1: SDE Diffusion



Exc. Kurtosis



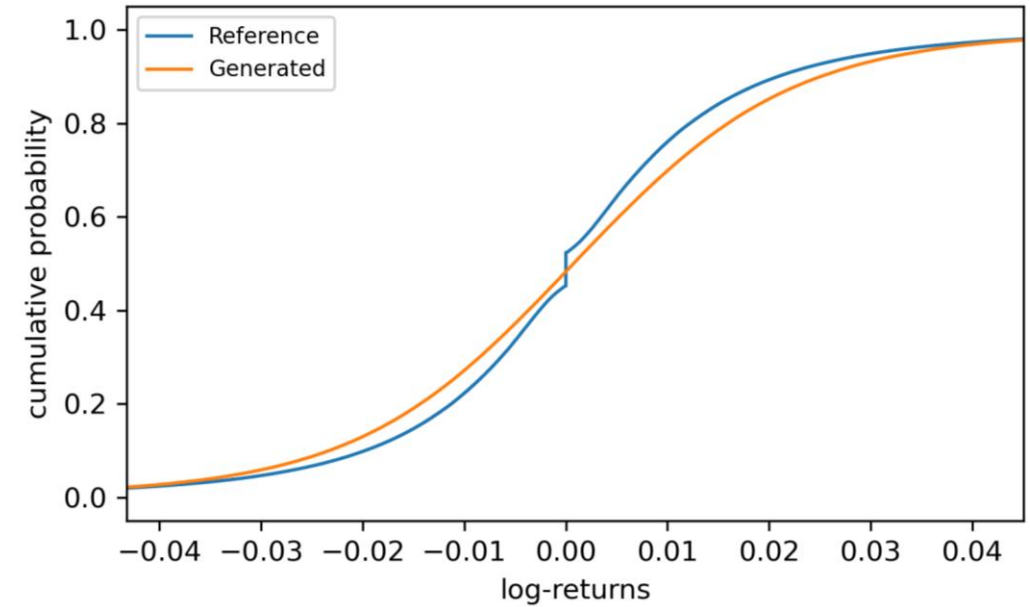
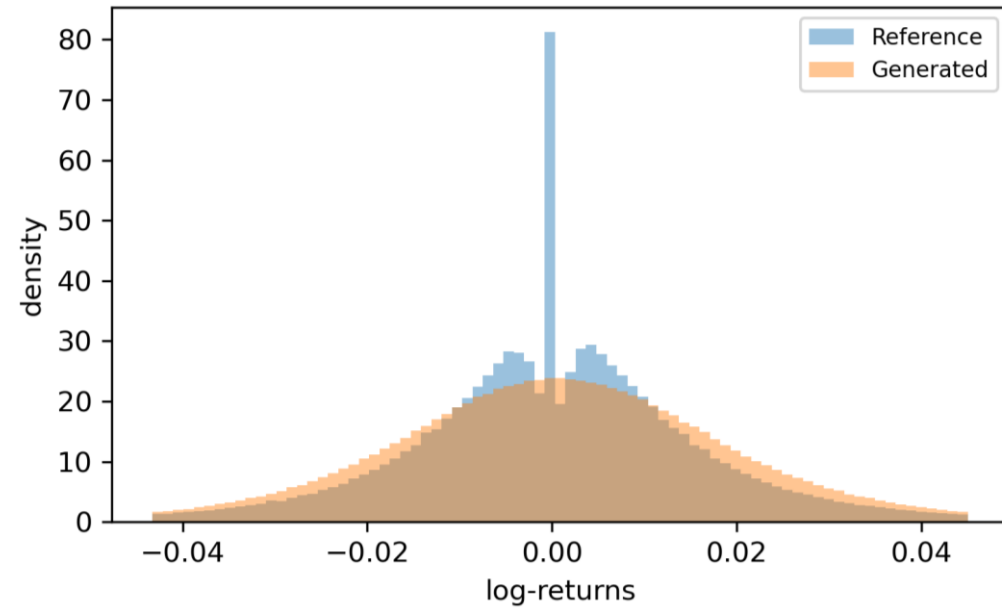
Leverage Effect



Zero-ret. Spike



Dispersion



Phase 1: SDE Diffusion



Exc. Kurtosis



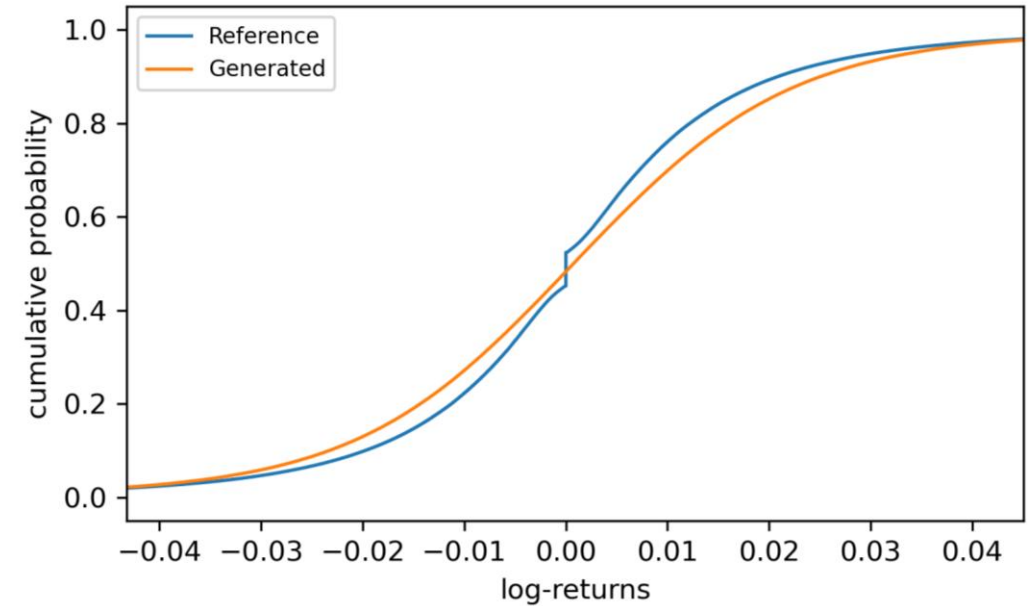
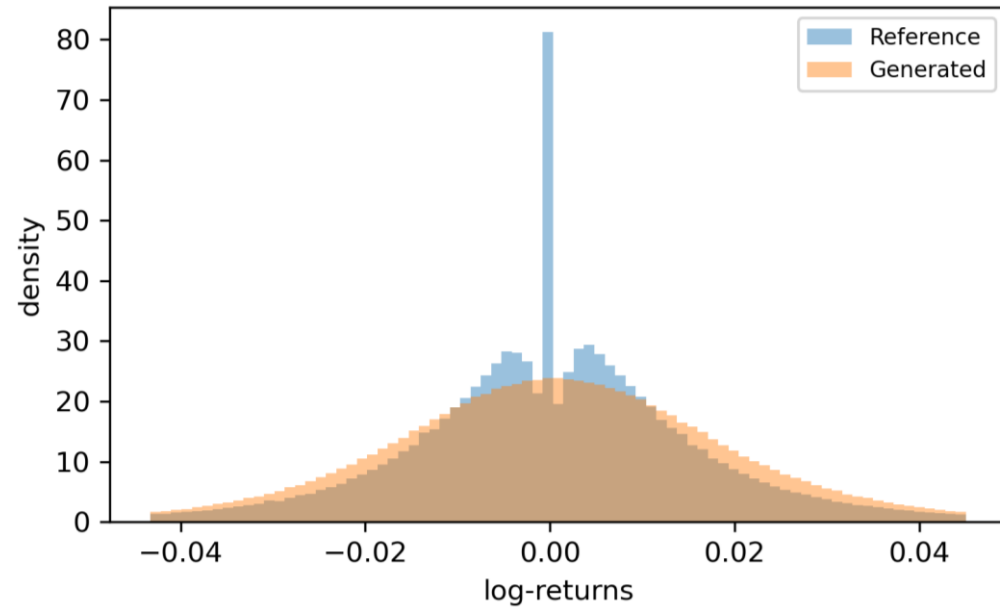
Leverage Effect



Zero-ret. Spike



Dispersion



VE: more stable behavior and best distributional fit

Phase 2: EDM-CSDI Unconditional



Exc. Kurtosis



Leverage Effect



Zero-ret. Spike



Dispersion



Leverage effect improves

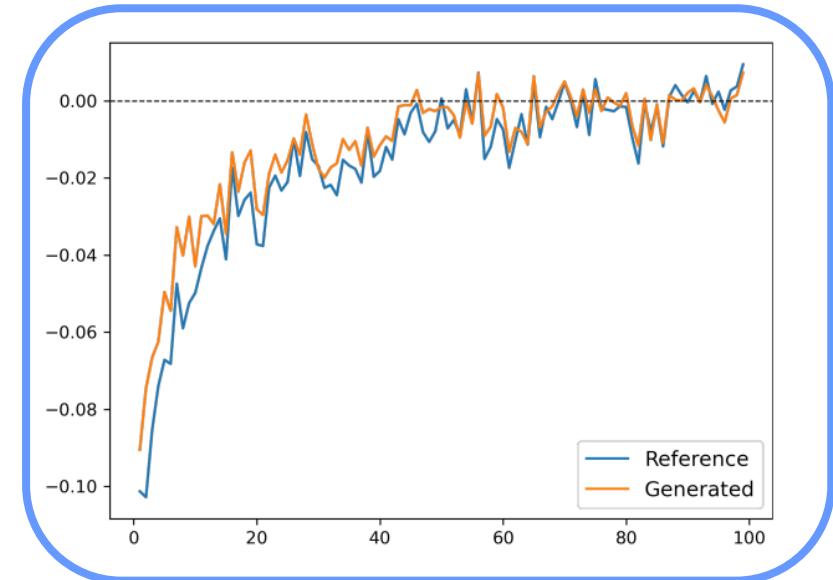
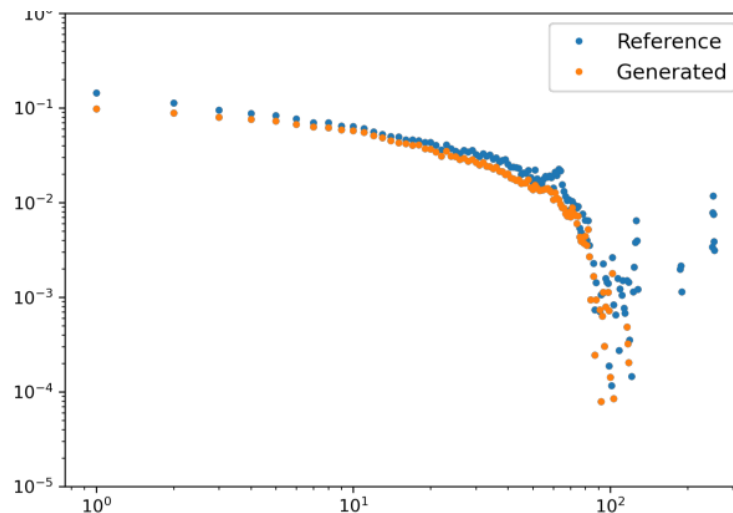
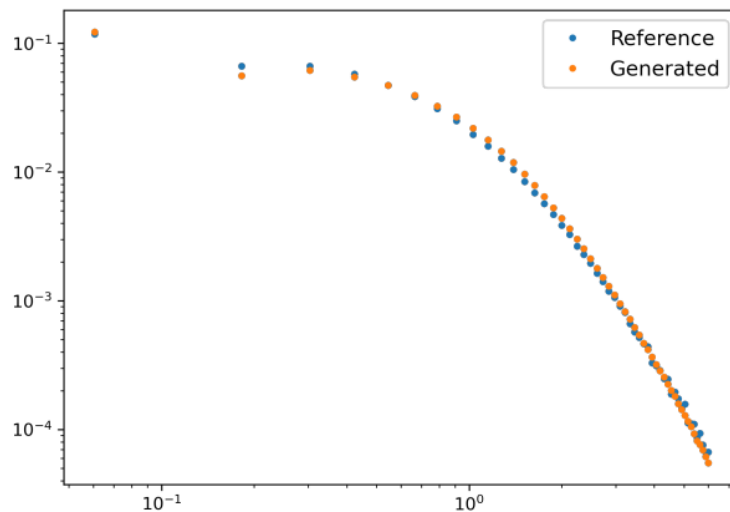


Std deteriorate



Kurtosis gap reduced
(60% → 49%)

Configuration	Mean	Std	Skew.	Kur.	$q_{0.01}$	$q_{0.99}$	$D_{KS} \downarrow$	$W_1 \downarrow$	$\hat{\alpha}$
Reference	0.0000	1.0000	-0.5864	39.5	-2.7188	2.7746	—	—	4.418
EDM-CSDI (Unc.)	0.0008	0.7523	-0.2292	20.0	-2.0292	2.0855	0.0418	0.0029	4.640



Phase 2: EDM-CSDI Unconditional



Exc. Kurtosis



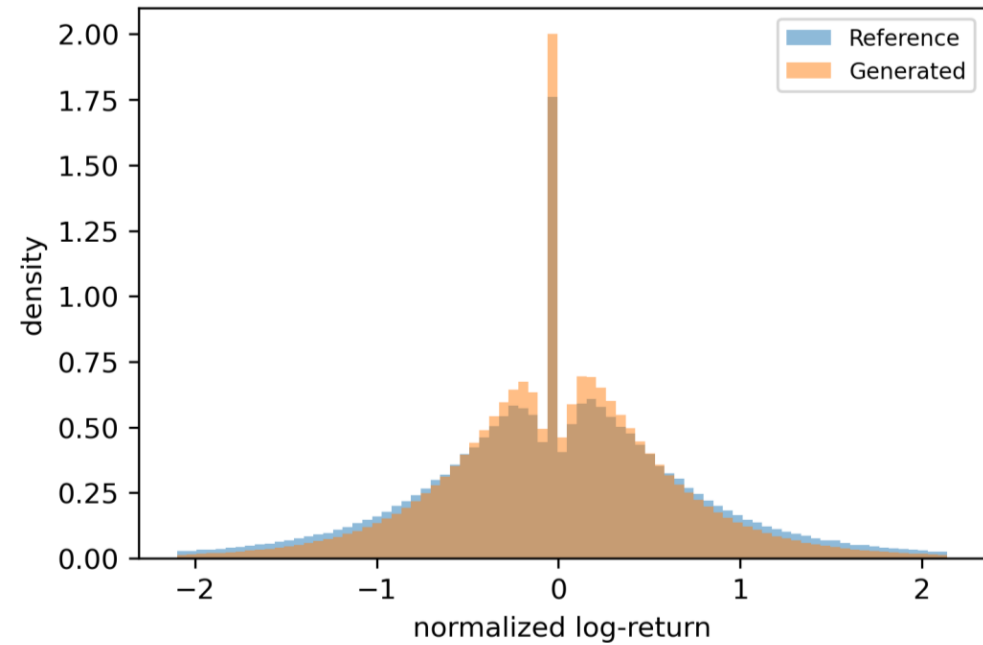
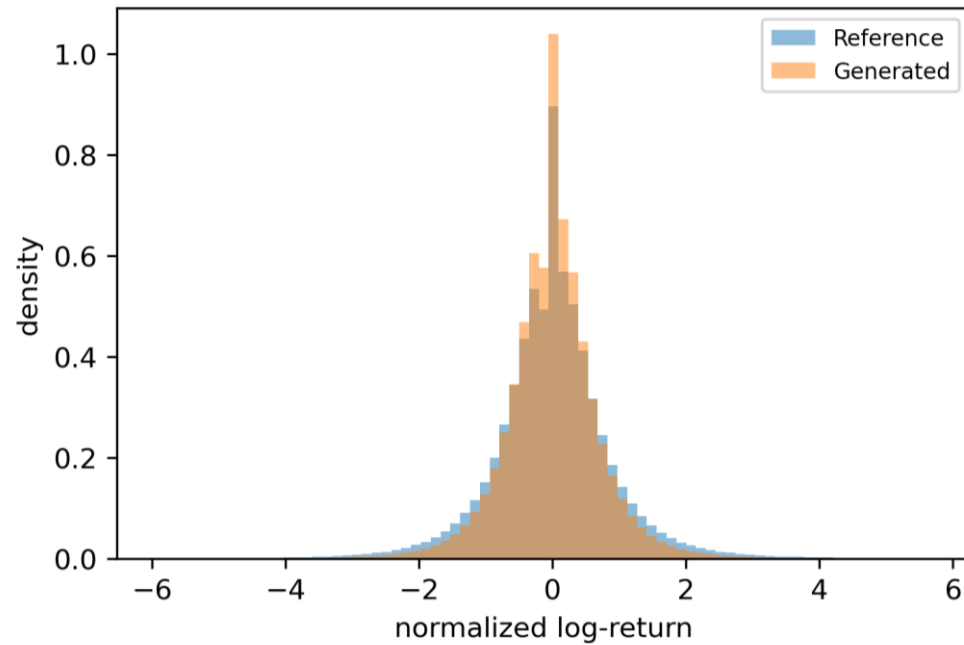
Leverage Effect



Zero-ret. Spike



Dispersion



Phase 2: EDM-CSDI Unconditional



Exc. Kurtosis



Leverage Effect

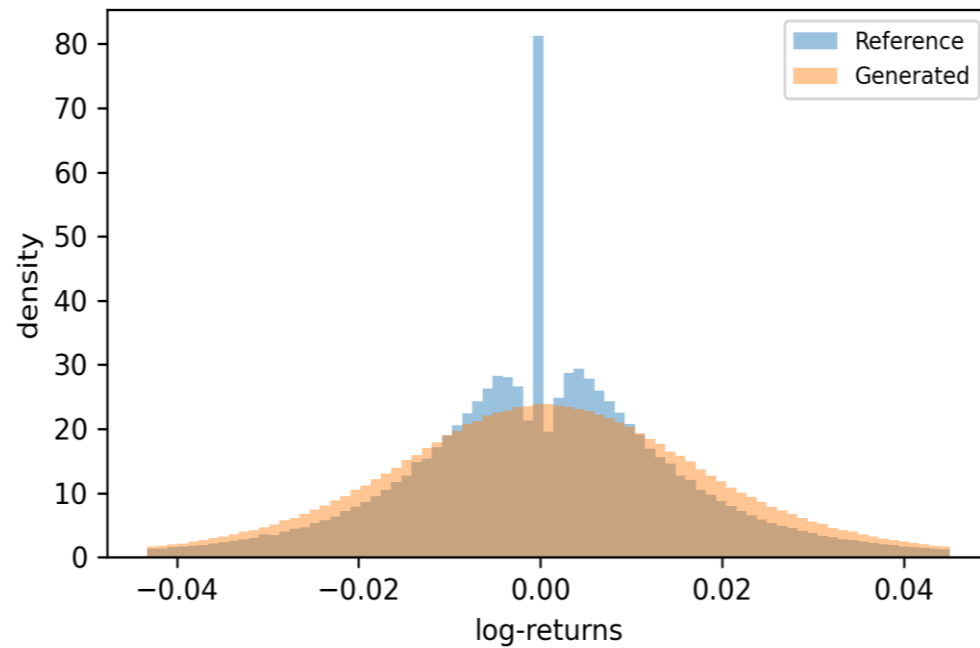


Zero-ret. Spike

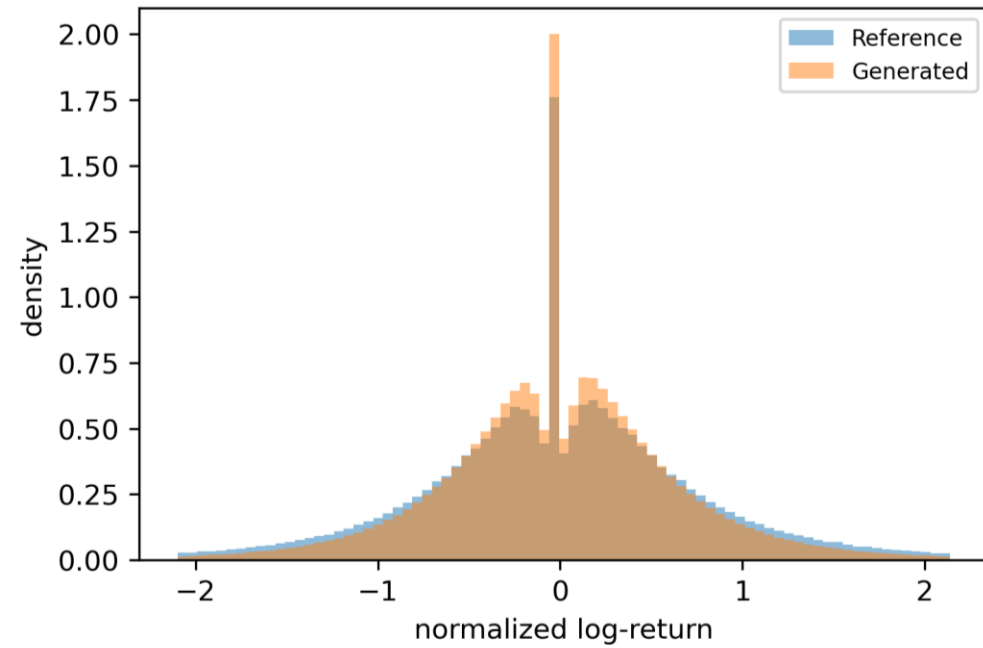


Dispersion

Phase 1 (VE+Exp.)



EDM-CSDI (Unc.)



EDM Set-Up + Change Objective

Phase 2: EDM-CSDI Conditional

Exc. Kurtosis

Leverage Effect

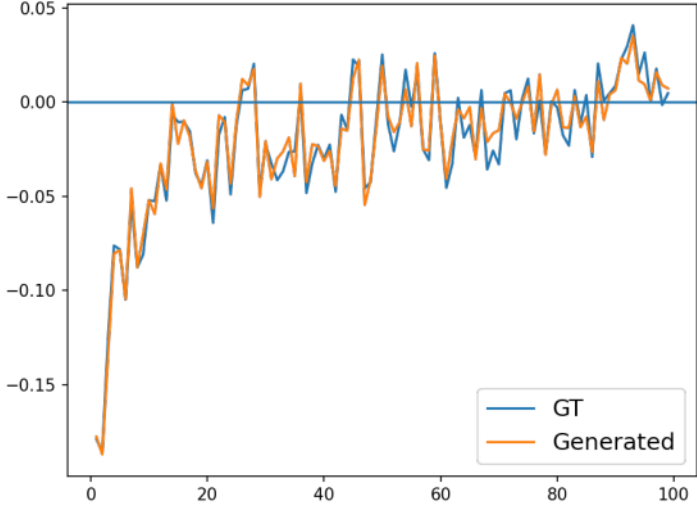
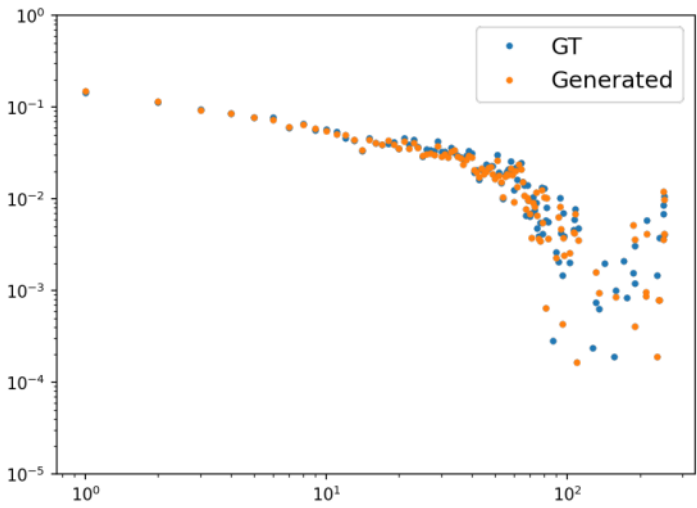
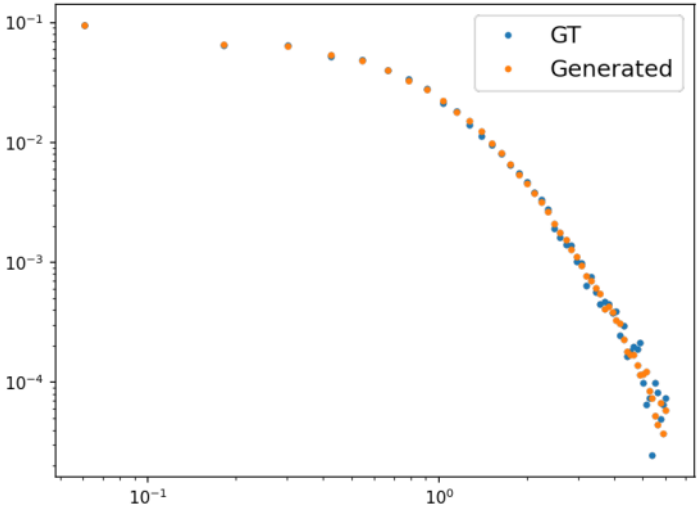
Zero-ret. Spike

Dispersion

- Lowest Wasserstein Dist.
- Std recovered (25% → 5%)
- Kurtosis train gap (49% → 1%)

Split	Configuration	Mean	Std	Skew.	Kur.	$q_{0.01}$	$q_{0.99}$	$D_{KS} \downarrow$	$W_1 \downarrow$	$\hat{\alpha}$
Val	Ground Truth	-0.0024	0.8656	-0.4069	13.9	-2.3305	2.3723	—	—	4.072
Val	EDM-CSDI (Unc.)	0.0031	0.7314	-0.3558	19.9	-1.9595	2.0232	0.0331	0.0020	4.768
Val	EDM-CSDI (Cond.)	0.0033	0.8299	-0.4432	13.9	-2.2396	2.2635	0.0352	0.0005	4.278
Both	Reference	0.0000	1.0000	-0.5864	39.5	-2.7188	2.7746	—	—	4.418
Both	EDM-CSDI (Unc.)	0.0008	0.7523	-0.2292	20.0	-2.0292	2.0855	0.0418	0.0029	4.640
Both	EDM-CSDI (Cond.)	0.0029	0.9487	-0.5390	39.0	-2.5789	2.6306	0.0411	0.0006	4.572

Validation Windows



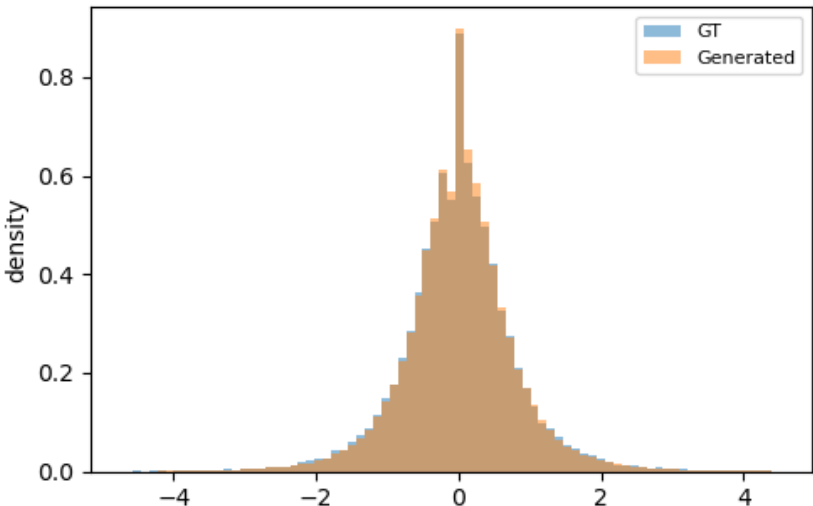
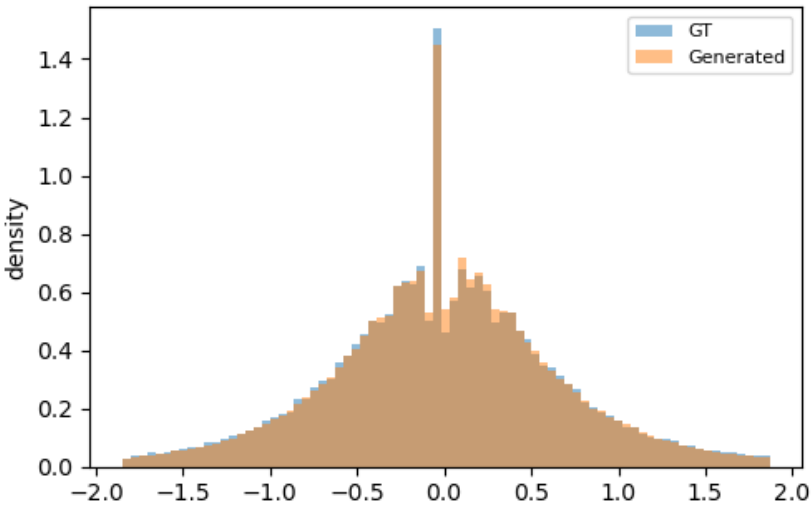
Phase 2: EDM-CSDI Conditional

Exc. Kurtosis

Leverage Effect

Zero-ret. Spike

Dispersion



- CRPS skill score
(27% Val, 40% Both)
- MAE fall
- Overconfident above 90%
(interval coverage)

Configuration	Split	N. Samples	MAE ↓	CRPS ↓	T_1^{norm}	T_1	T_2^{norm}
Climatology	Val	50	0.60851	0.44900	0.89782	0.01873	0.89765
OLS	Val	50	0.25362	0.18690	0.37385	0.00780	0.37390
EDM-CSDI (Cond.)	Val	50	0.18441	0.12141	0.24431	0.00510	0.24584

Table 6.6

Phase 2: EDM-CSDI Conditional

Exc. Kurtosis

Leverage Effect

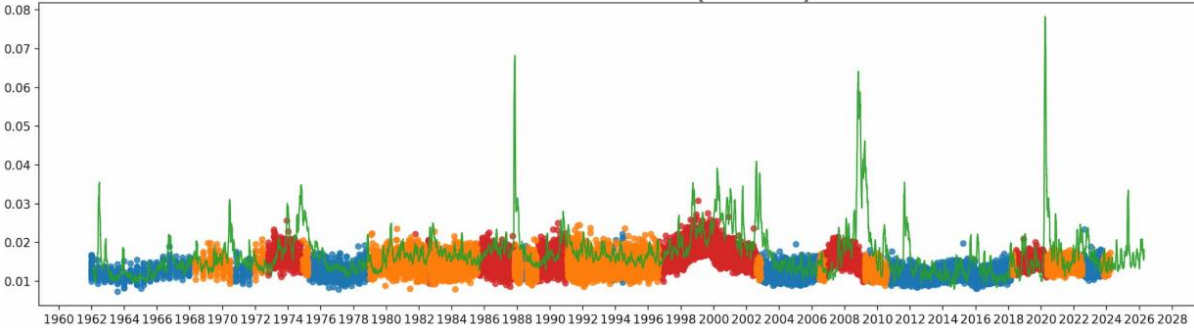
Zero-ret. Spike

Dispersion

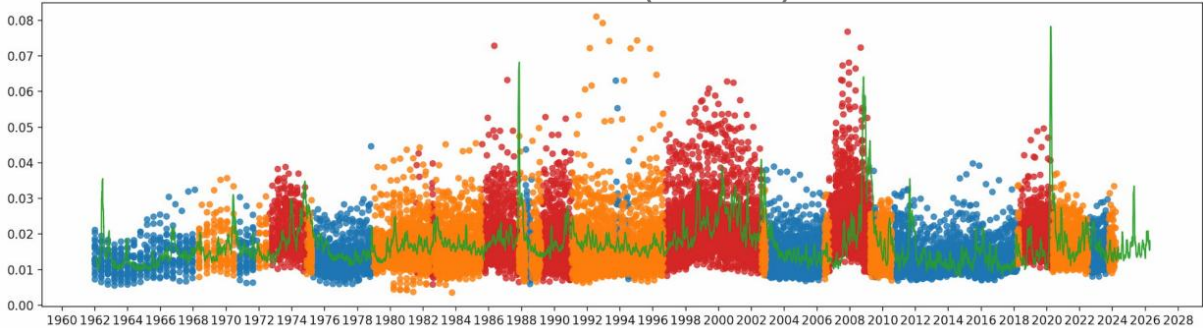
- EDM-CSDI Cond. top performance in High
- EDM-CSDI Unc. averaged dispersion levels

Gen/Ref ratio	Low	Medium	High
EDM-CSDI (Unc.)	0.87	0.81	0.74
EDM-CSDI (Cond.)	0.95	0.94	0.99

EDM-CSDI (Unc.)



EDM-CSDI (Cond.)



Phase 2: EDM-CSDI Conditional

Exc. Kurtosis

Leverage Effect

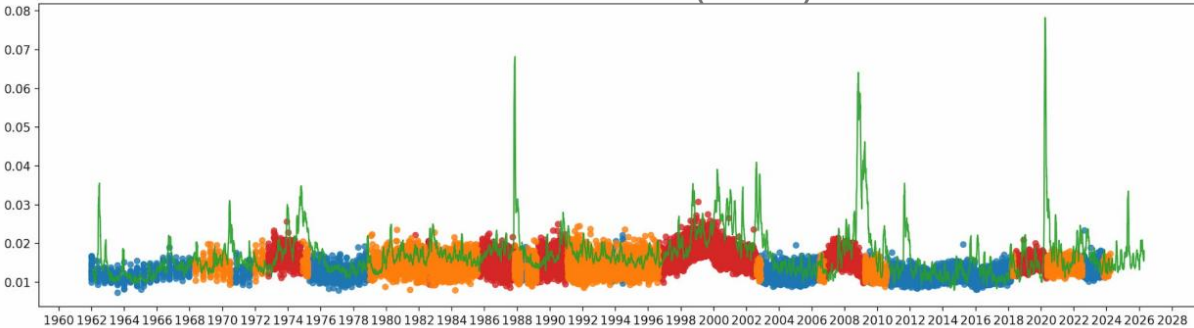
Zero-ret. Spike

Dispersion

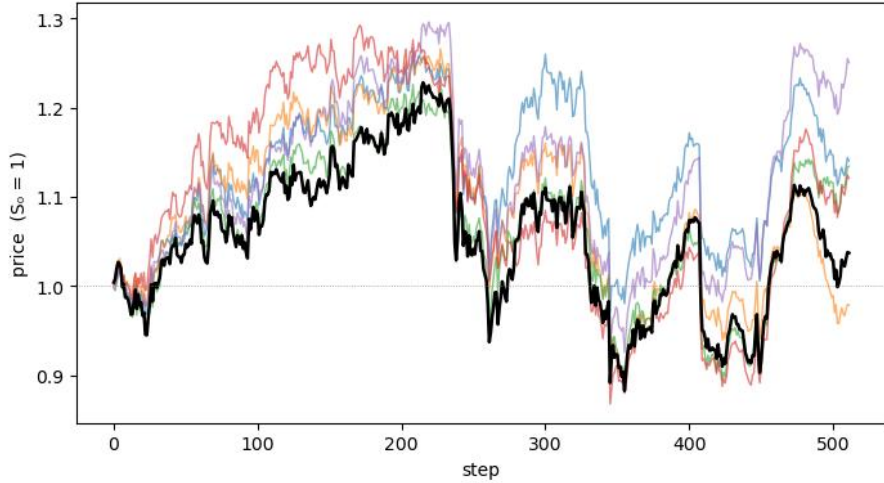
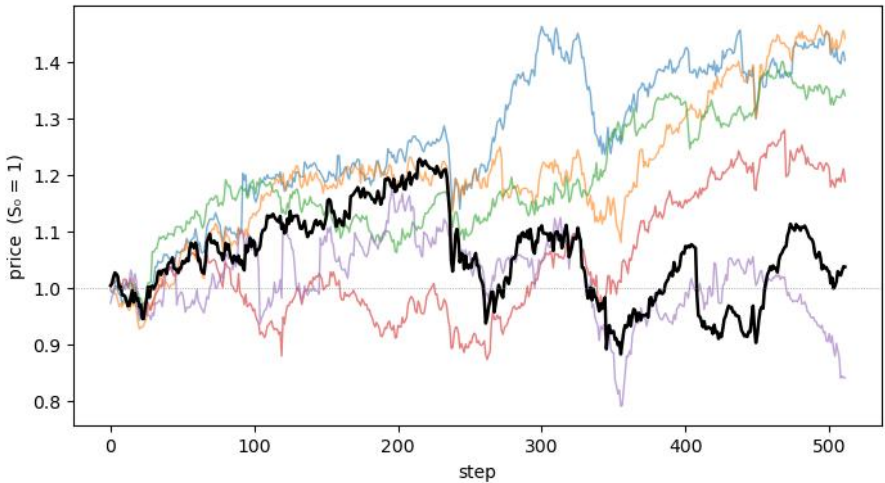
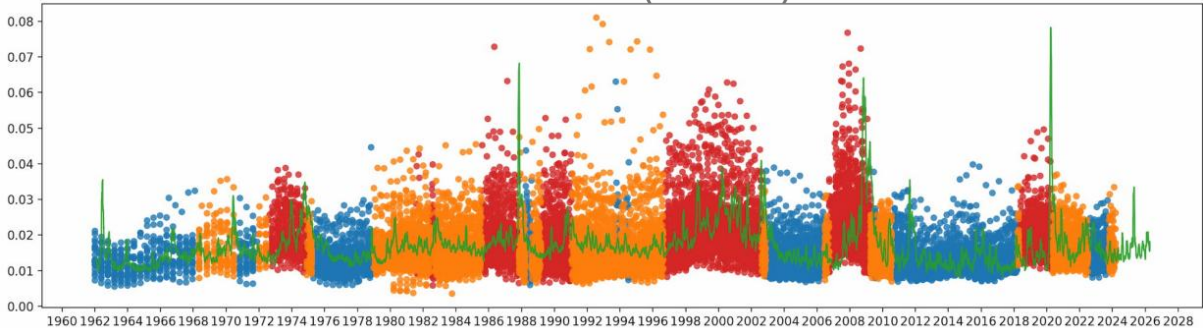
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EDM-CSDI (Unc.)



EDM-CSDI (Cond.)

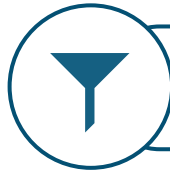


Limitations and Future Work

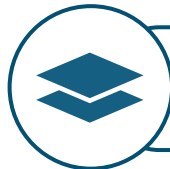
Limitations



EDM and objective confounded



Small forecast sampling budget



No deep generative baselines

Limitations and Future Work

Limitations

-  EDM and objective confounded
-  Small forecast sampling budget
-  No deep generative baselines

Future Work

-  Heavy-tail forward noising
-  Add financial risk metrics
-  Regime-stratified reweighting

Thank you.

Questions?

Appendix

Additional Experiments

- Heun SDE
 - [Result Tables & Stylized Facts](#)
 - [EDM-CSDI CRPS & Coverage Interval](#)
- Volatility Analysis
 - [Mean Volatility Proxy Table](#)
- [Capacity Ablation and analysis](#)

Results Tables and Images

- [Phase 1](#)
 - [Stylized Facts: Heavy Tails](#)
 - [Stylized Facts: Volatility Clustering](#)
 - [Stylized Facts: Leverage Effect](#)
- Phase 2
 - [EDM-CSDI CRPS](#)
 - [EDM-CSDI Coverage Interval](#)
 - [EDM-CSDI Price Paths Comparison](#)
 - [EDM-CSDI Price Paths Aggregated](#)

Architectures and Training parameters

- [CSDI Architecture](#)
 - [CSDI Conditional Mechanism](#)
- [Kim et al. Architecture](#)
- [EDM-CSDI Architecture](#)
 - [EDM-CSDI Architecture: ResidualBlock](#)
- [Validation Window Construction](#)
- [Noise Schedulers](#)

Overview Tables

- [Phase 1 vs Phase 2](#)
- [Training Setup](#)

Limitations and Future Work

- [Limitations](#)
- [Future Work](#)

Phase 1 vs Phase 2

Category	Dimension	Phase 1 — Replication	Phase 2 — EDM-CSDI	Category	Dimension	Phase 1 — Replication	Phase 2 — EDM-CSDI
<i>Problem setup</i>	Primary objective	Replicate Kim et al.[17]; validate VE, VP, and GBM-inspired variants on unconditional univariate generation	Conditional generation of Close log-returns given observed O, H, L; CSDI backbone integrated within the EDM framework	<i>Training objective</i>	Prediction target	Noise ϵ (DSM)	Clean data x_0 (MMSE)
	Channels K	$K = 1$ (univariate close log-returns)	$K = 4$ (OHLC; all channels as log-returns relative to C_{t-1})		Loss function	$\mathbb{E}[\ \epsilon - \epsilon_\theta(x_t, t)\ _{m^{ta}}^2]$	$\mathbb{E}[\lambda(\sigma)\ D_\theta(x_t; \sigma) - x_0\ _{m^{ta}}^2]$
	Conditioning / masking	unconditional : $m^{co} = 0$; loss computed on all entries, no side information	predict_close : $\{O, H, L\}$ fully observed, $\{C\}$ fully masked; fixed deterministic split		Loss weighting λ	$\lambda(\sigma_i) = \sigma_i^2$	$\lambda(\sigma) = c_{out}(\sigma)^{-2}$
	Sequence length / stride	$L = 2048, s = 400$	$L = 512, s = 100$		Preconditioning	None; raw network $F_\theta(x_t; t)$	$D_\theta = c_{skip} x + c_{out} F_\theta(c_{in} x; c_{noise})$, with $\sigma_{data}=1.0$
<i>Diffusion process</i>	Forward SDE family	VE; VP; GBM-inspired VE in log-price space	VE-style forward process within EDM preconditioning	<i>Sampling</i>	Diffusion-step embedding	Discrete step index t (sinusoidal basis + two linear layers with SiLU)	$c_{noise}(\sigma) = \frac{1}{4} \ln \sigma$; compresses the dynamic range of σ to a bounded scalar
	σ_{data}	Implicit; absorbed into the choice of $\sigma_{max}/\sigma_{min}$	$\sigma_{data} = 1.0$ by construction (per-channel z -score normalisation)		Numerical sampler	Probability flow ODE (first order)	Probability flow ODE (second-order Heun)
	Noise schedule parameters	VE: $\sigma_{max}=1, \sigma_{min}=0.01$; GBM: $\sigma_{max}=10, \sigma_{min}=0.1$; VP: $\beta_{max}=8, \beta_{min}=0.01$	$P_{mean}=-1.4, P_{std}=1.8$ (ablation-tuned)		Sampler-training coupling	Coupled	Decoupled
	Noise level sampling	$t \sim \mathcal{U}[0, 1]$	$\ln \sigma \sim \mathcal{N}(P_{mean}, P_{std}^2)$				

Training Setup

Category	Parameter	Phase 1 — Replication	Phase 2 — EDM-CSDI	Category	Parameter	Phase 1 — Replication	Phase 2 — EDM-CSDI
<i>Data & pre-processing</i>	Channels K	$K = 1$ (close log-return)	$K = 4$ (OHLC)	<i>Noise schedule</i>	$\sigma_{\max} / \sigma_{\min}$	VE: $\sigma_{\max}=1$, $\sigma_{\min}=0.01$; GBM: $\sigma_{\max}=10$, $\sigma_{\min}=0.1$	—
	Preprocessing	Raw close log-return	OHLC channels as log-returns relative to C_{t-1} ; per-channel z -score normalisation		$\beta_{\max} / \beta_{\min}$	VP: $\beta_{\max}=8$, $\beta_{\min}=0.01$	—
					$P_{\text{mean}} / P_{\text{std}}$	—	$P_{\text{mean}}=-1.4$, $P_{\text{std}}=1.8$
	Conditioning	Unconditional; $m^{co} = \mathbf{0}$	predict_close: $\{O, H, L\}$ observed, $\{C\}$ masked	<i>Training</i>	Validation wind.	Partly overlapping	Fully held-out
	Window length L	2048 (≈ 8 years)	512 (≈ 2 years)		LR schedule	cosine annealing	
<i>Architecture</i>	Stride s	400	100		$\eta_{\max} / \eta_{\min}$	$5 \times 10^{-4} / 1 \times 10^{-6}$	$1 \times 10^{-4} / 1 \times 10^{-6}$
	Model channels		128		Epoch	1000	500
	Layers		6		Batch size		64
	Diffusion emb.		256		Optimizer		AdamW
	Time emb.		128		Early-stopping	—	validation-loss; 200 epochs
	Feature emb.		64	<i>Sampling</i>	Method	Euler–Maruyama SDE Probability-flow ODE	Heun ODE Stochastic Heun SDE
					$\sigma_{\max} / \sigma_{\min}$	from <i>Noise schedule</i>	$\sigma_{\max} = 8.0$, $\sigma_{\min} = 0.002$

Limitations and Future Work

Limitations

-  No extensive hyperparameter ablation
-  EDM and conditioning confounded
-  Capacity vs. dataset size
-  Small forecast sampling budget
-  No deep generative baselines

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Future Work

-  Random Conditioning Mask
-  Heavy-tail forward noising
-  Add financial risk metrics
-  Regime-stratified reweighting
-  Extend Heun SDE sampling exploration

Additional Results: Heun SDE

- Best distributional fit
- Std increase, closer to Ref.
- Kurtosis exceed ref
- Similar Stylized Facts

Sampler	Mean	Std	Skew.	Kur.	$q_{0.01}$	$q_{0.99}$
Reference	0.0000	1.0000	-0.5864	39.5	-2.7188	2.7746
Heun ODE	0.0029	0.9487	-0.5390	39.0	-2.5789	2.6306
Heun SDE	0.0012	1.0041	-0.6876	45.6	-2.7224	2.7662

Sampler	D_{KS}	W_1	$\hat{\alpha}$
Reference	—	—	4.418
Heun ODE	0.0411	0.000569	4.572
Heun SDE	0.0377	0.000077	4.599

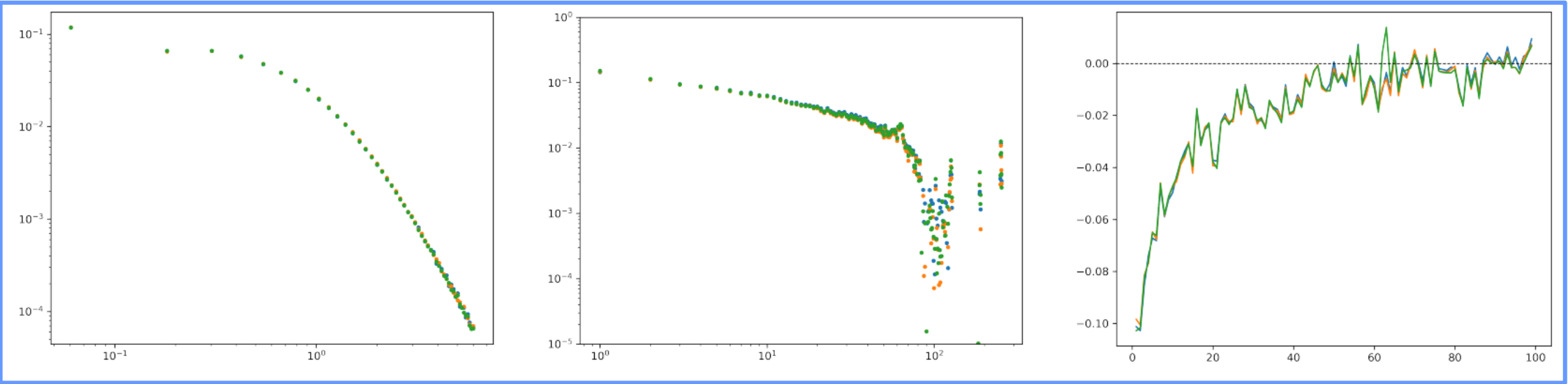


Table C.1, Table C.2, and Figure C.1

Additional Results: Heun SDE



Marginally better CRPS



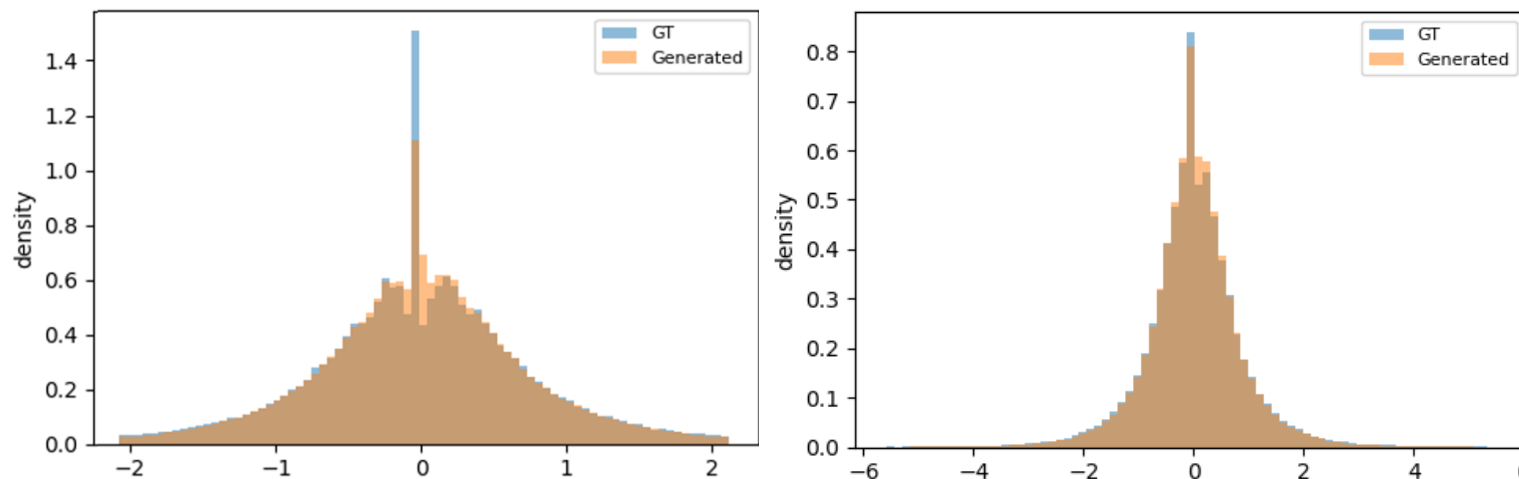
Similar coverage performance

Split	Nominal (%)	EDM-CSDI 25, Heun ODE	EDM-CSDI 25, Heun SDE
Val	50.0	50.59	50.95
Val	80.0	78.41	78.41
Val	90.0	86.66	86.47
Val	95.0	91.34	91.06
Val	99.0	93.46	93.23
Both	50.0	51.57	51.53
Both	80.0	78.95	78.75
Both	90.0	86.87	86.58
Both	95.0	91.46	91.11
Both	99.0	93.60	93.26

Configuration	Split	Sampler	N. Samples	MAE ↓	CRPS ↓	T_1^{norm}	T_1	T_2^{norm}
Climatology	Val	—	25	0.619 13	0.448 69	0.897 45	0.018 73	0.897 53
EDM-CSDI (Cond.)	Val	Heun ODE	25	0.186 00	0.121 44	0.244 38	0.005 10	0.245 88
EDM-CSDI (Cond.)	Val	Heun SDE	25	0.181 00	0.120 40	0.238 33	0.004 97	0.235 87
Climatology	Both	—	25	0.662 88	0.482 32	0.964 66	0.020 13	0.964 29
EDM-CSDI (Cond.)	Both	Heun ODE	25	0.195 46	0.126 02	0.257 31	0.005 37	0.262 58
EDM-CSDI (Cond.)	Both	Heun SDE	25	0.188 75	0.123 89	0.249 81	0.005 21	0.251 83

Additional Results: Capacity Ablation and Analysis

- Distributional properties retained
- Mixed $\hat{\alpha}$ results
- Kurtosis Blow-ups
- Std variation



Split	Configuration	Mean	Std	Skew.	Kur.	$q_{0.01}$	$q_{0.99}$	$D_{KS} \downarrow$	$W_1 \downarrow$	$\hat{\alpha}$
Val	Ground Truth	-0.0024	0.8656	-0.4069	13.9	-2.3305	2.3723	—	—	4.072
Val	EDM-CSDI (Cond.), 128 ch	0.0033	0.8299	-0.4432	13.9	-2.2396	2.2635	0.0352	0.0005	4.278
Val	EDM-CSDI (Cond.), 64 ch	0.0071	0.9190	-0.1553	21.4	-2.5076	2.5359	0.0425	0.0007	4.110
Both	Reference	0.0000	1.0000	-0.5864	39.5	-2.7188	2.7746	—	—	4.418
Both	EDM-CSDI (Cond.), 128 ch	0.0029	0.9487	-0.5390	39.0	-2.5789	2.6306	0.0411	0.0006	4.572
Both	EDM-CSDI (Cond.), 64 ch	0.0056	0.9394	-0.0024	114.4	-2.5408	2.5999	0.0390	0.0006	4.936

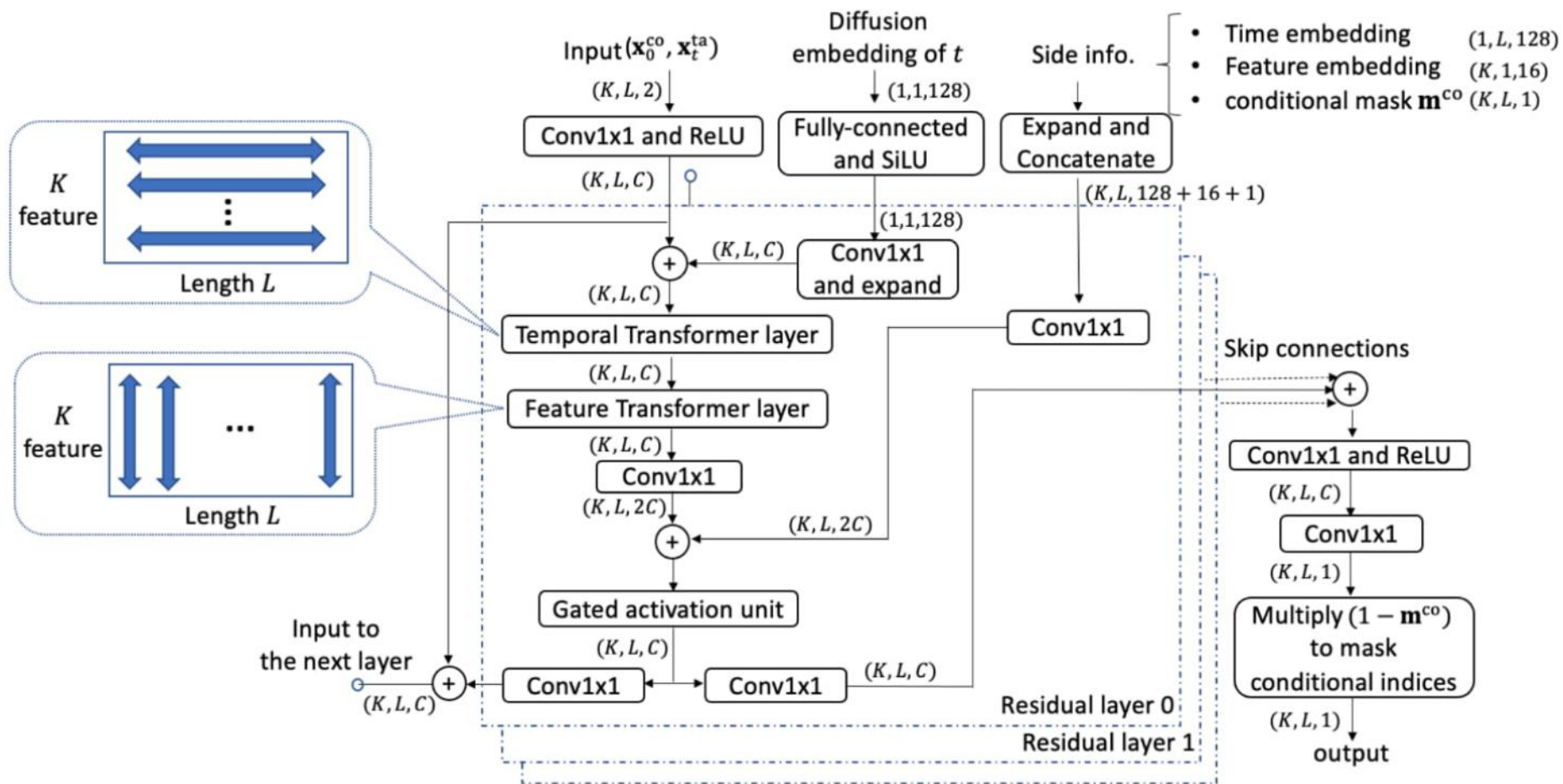
Additional Results: Capacity Ablation and Analysis

-  Slightly worse CRPS
-  Better coverage at high %
-  Higher underconfidence at Low %

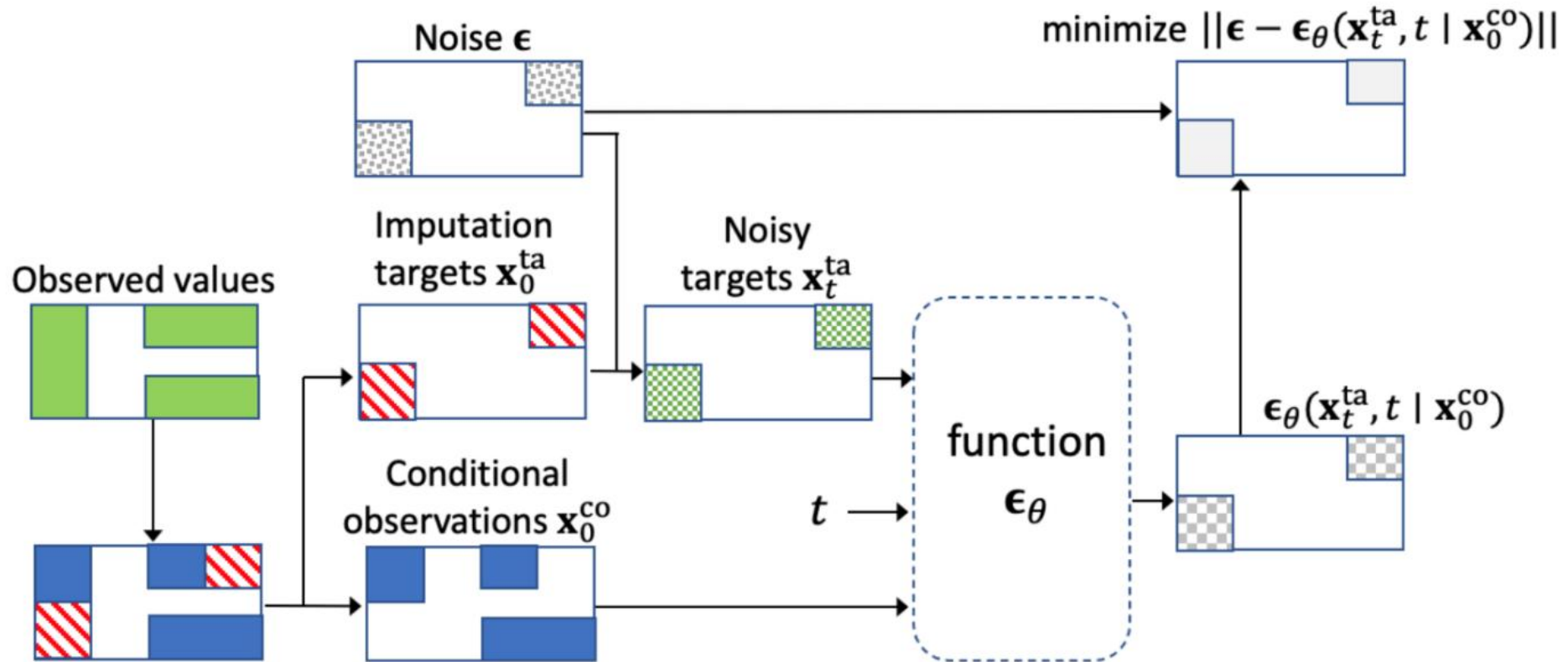
Split	Nominal (%)	EDM-CSDI 50, 128 ch	EDM-CSDI 50, 64 ch
Val	50.0	52.23	54.32
Val	80.0	82.16	83.91
Val	90.0	90.35	91.75
Val	95.0	94.06	95.11
Val	99.0	96.89	97.61
Both	50.0	53.05	54.76
Both	80.0	82.52	84.04
Both	90.0	90.50	91.55
Both	95.0	94.02	94.78
Both	99.0	96.84	97.26

Configuration	Split	N. Samples	MAE ↓	CRPS ↓	T_1^{norm}	T_1	T_2^{norm}
EDM-CSDI (Cond.), 128 ch	Val	50	0.184 41	0.121 41	0.244 31	0.005 10	0.245 84
EDM-CSDI (Cond.), 64 ch	Val	50	0.220 41	0.145 08	0.292 99	0.006 11	0.295 82
EDM-CSDI (Cond.), 128 ch	Both	50	0.193 79	0.126 59	0.257 27	0.005 37	0.262 59
EDM-CSDI (Cond.), 64 ch	Both	50	0.216 96	0.142 32	0.289 71	0.006 05	0.294 79

CSDI Architecture



CSDI Conditional Mechanism



Kim et al. Architecture

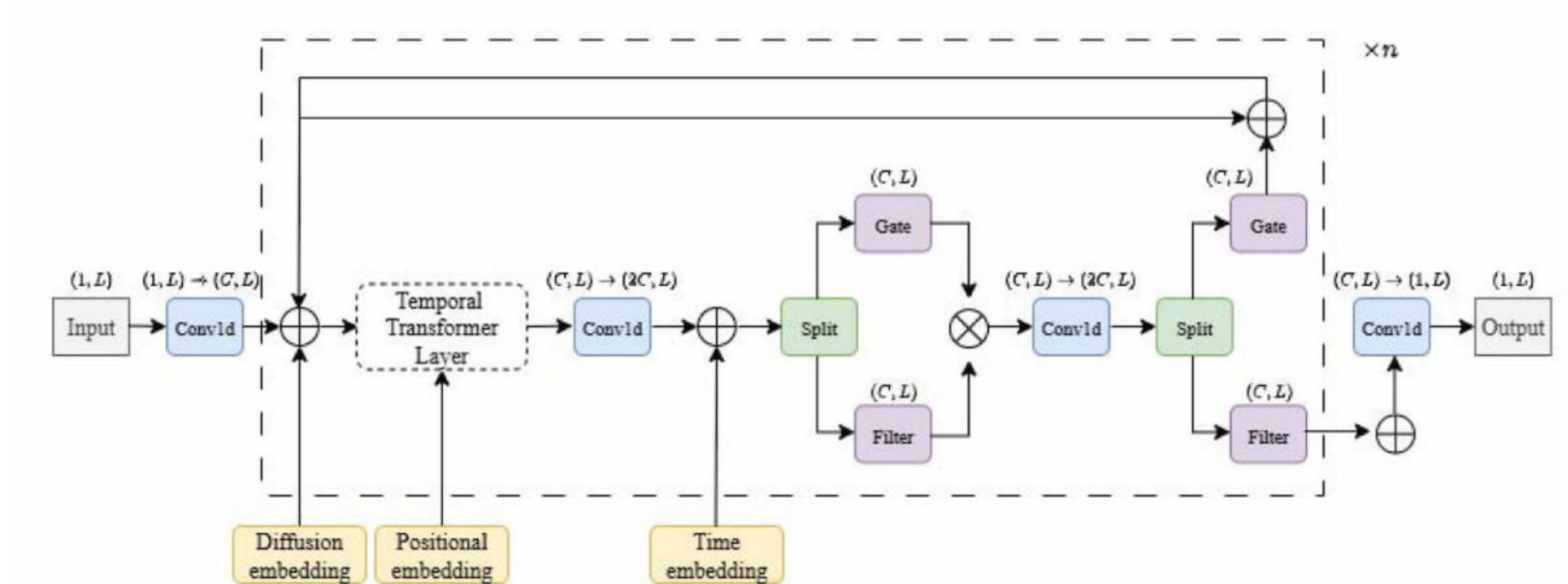


Figure 3.1: CSDI-based diffusion architecture. Source: Kim et al. [17].

EDM-CSDI Architecture

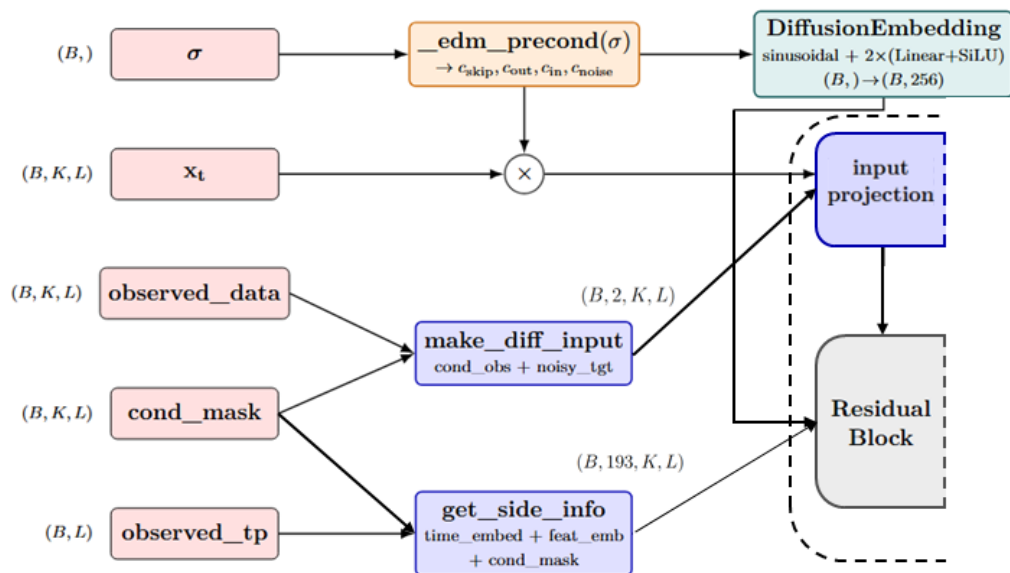
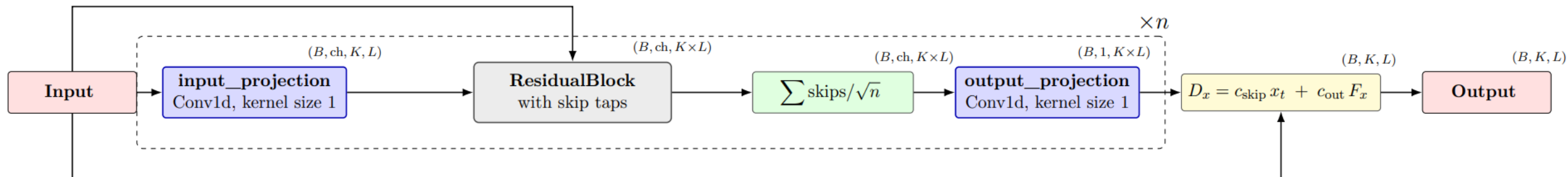


Figure 3.5: Representation of model inputs and how they are fed into the Denoiser. **input projection** and **ResidualBlock** are those represented in Fig. 3.3.

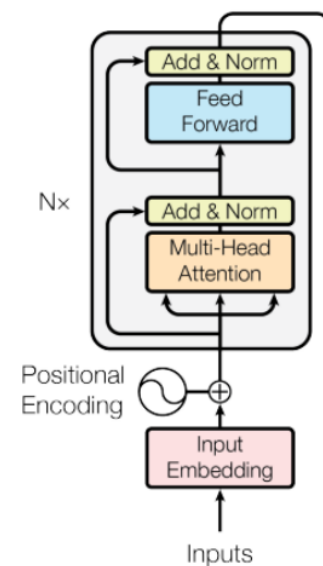


Figure 3.6: Transformer model architecture for the Encoder. Source: Vaswani et al. [38].

EDM-CSDI Architecture: ResidualBlock

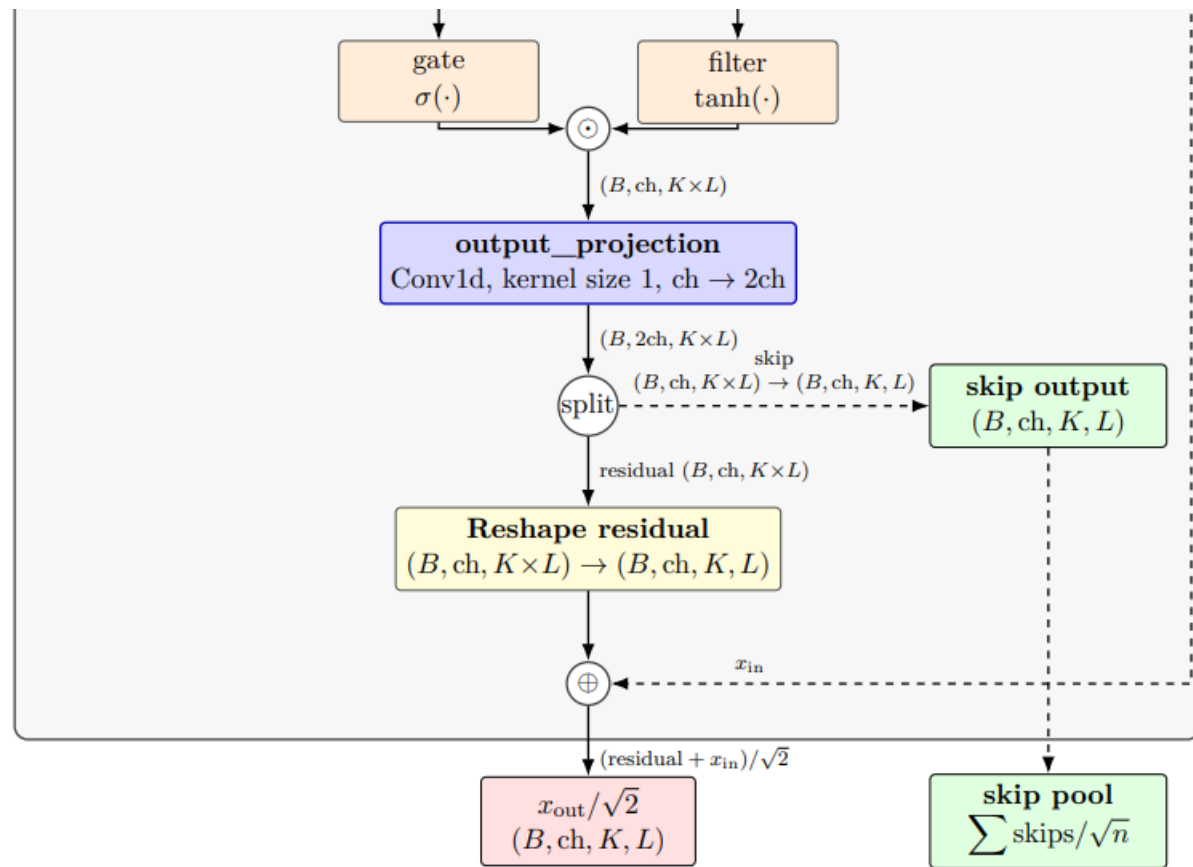
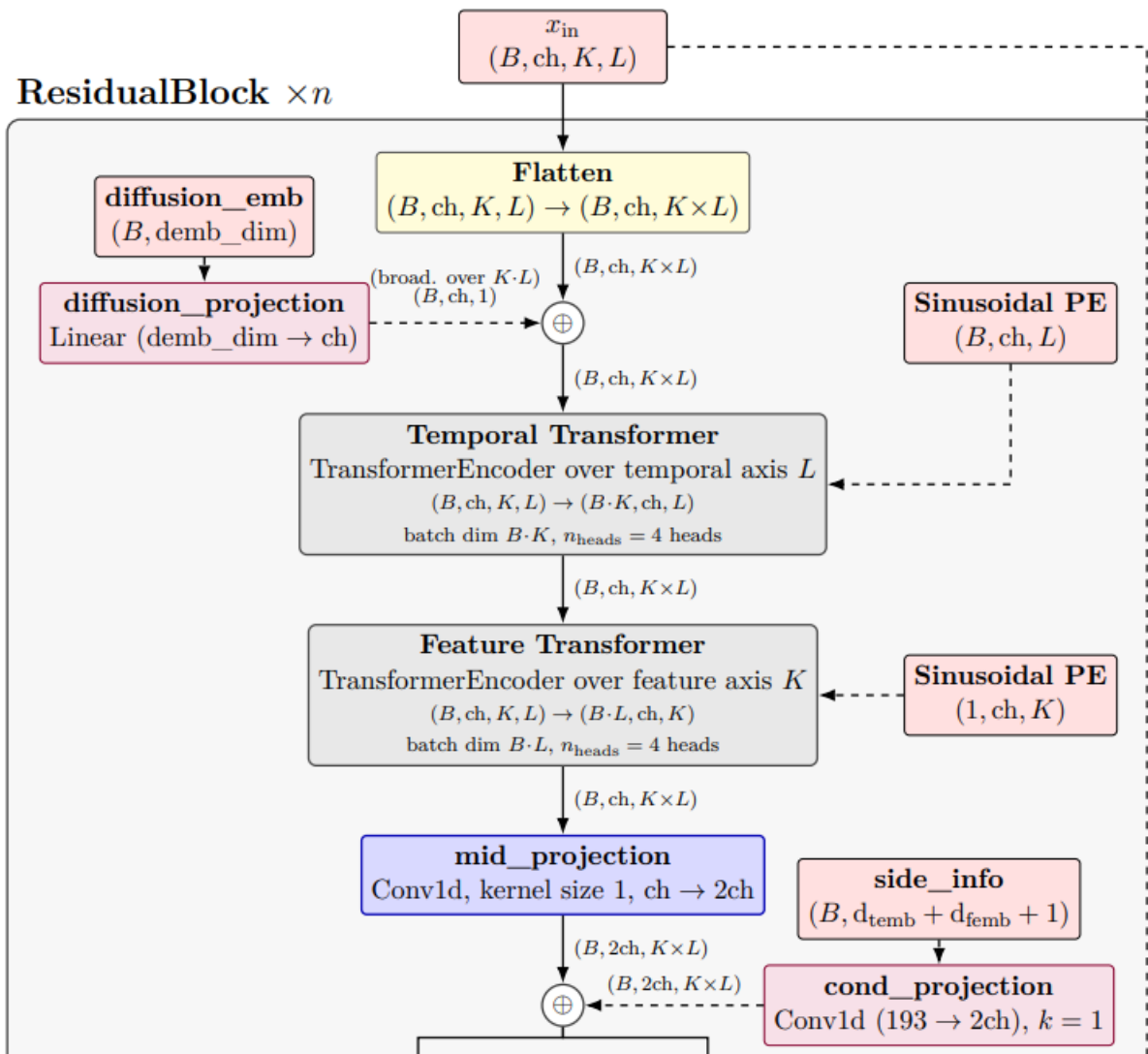


Figure 3.4: ResidualBlock architecture, CSDI-inspired. Each block emits a skip connection aggregated across all n_{layers} blocks (see Fig. 3.3).

Validation Window Construction

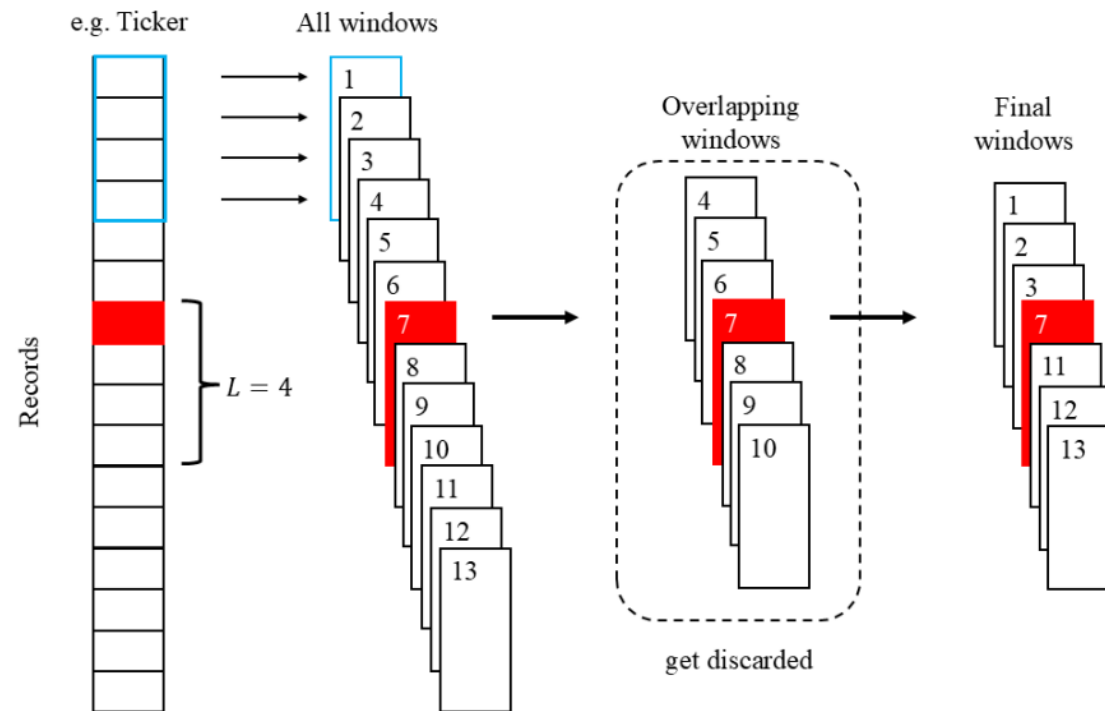
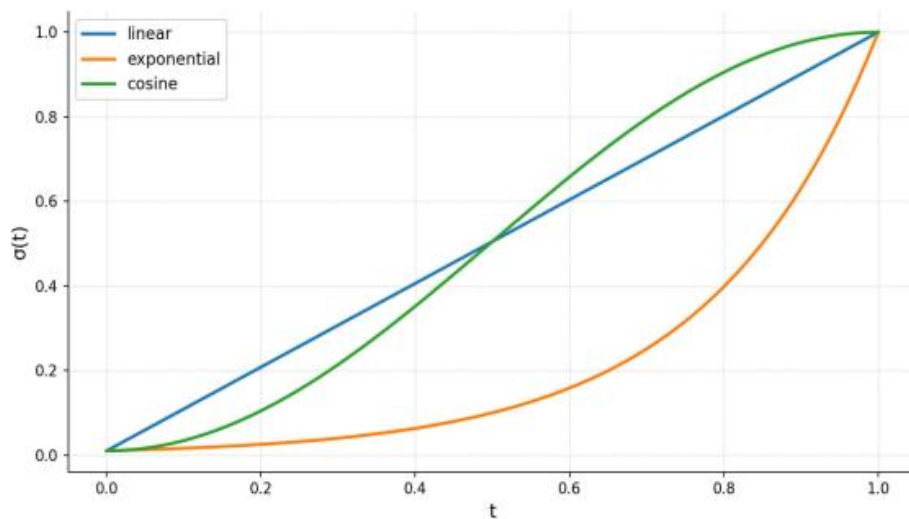
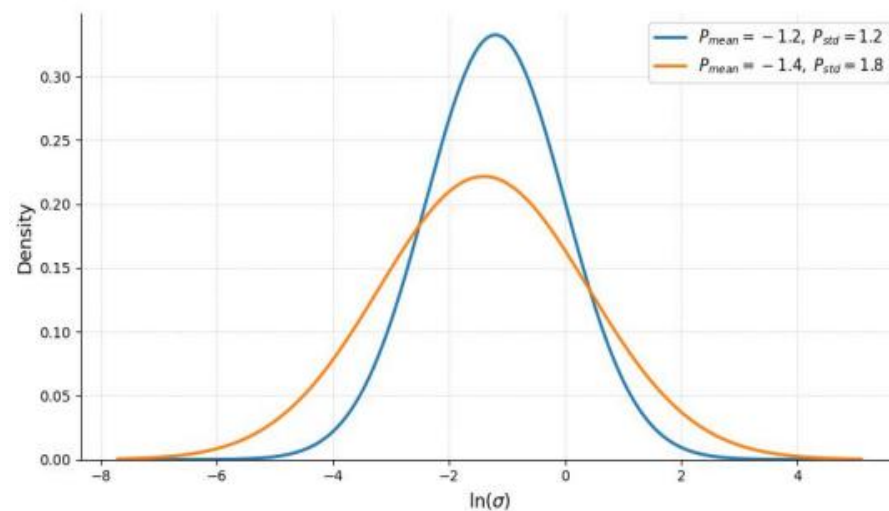


Figure 4.1: Phase 2 selection of Validation windows process. Example with window length $L = 4$, stride $s = 1$, and `val_split_ratio`= 0.25, where the 7th ticker index (red block) is drawn at random.

Noise Schedulers



(a) Sigma Schedule ($\sigma_{\min} = 0.01$, $\sigma_{\max} = 1.0$)
for $t \sim \mathcal{U}(0, 1]$ and noise $\sigma(t)$



(b) $\ln(\sigma) \sim \mathcal{N}(P_{\text{mean}}, P_{\text{std}}^2)$

Phase 1

Configuration	Mean	Std	Skew.	Kur.	$q_{0.01}$	$q_{0.99}$	$D_{\text{KS}} \downarrow$	$W_1 \downarrow$	$\hat{\alpha}$
Reference	0.0005	0.0208	-0.5387	36.0	-0.0562	0.0583	—	—	4.378
VE + Linear	0.0004	0.0270	-0.1712	7.0	-0.0649	0.0659	0.121	0.007	5.429
VE + Exponential	0.0005	0.0190	-0.0860	14.3	-0.0479	0.0492	0.037	0.002	4.461
VE + Cosine	0.0008	0.0221	-0.5979	21.4	-0.0558	0.0575	0.062	0.003	4.448
VP + Linear	0.0023	0.0446	6.3270	529.8	-0.0739	0.0891	0.142	0.011	4.883
VP + Exponential	0.0011	0.0546	0.1530	5.6	-0.1365	0.1539	0.206	0.025	5.535
VP + Cosine	0.0003	0.0235	-0.2985	15.7	-0.0607	0.0627	0.060	0.003	4.368
GBM + Linear	0.0001	0.2797	-0.0023	0.0	-0.6519	0.6516	0.419	0.210	13.055
GBM + Exponential	0.0001	0.1453	-0.0122	0.0	-0.3385	0.3379	0.370	0.101	10.935
GBM + Cosine	0.0000	0.1437	-0.0130	0.0	-0.3350	0.3342	0.371	0.102	11.134

Table 6.2: Aggregate distributional statistics for Phase 1 generated close log-returns and the training reference, computed over pooled window-level returns ($L = 2048$, stride=400).

Stylized Facts Phase 1: Heavy Tails

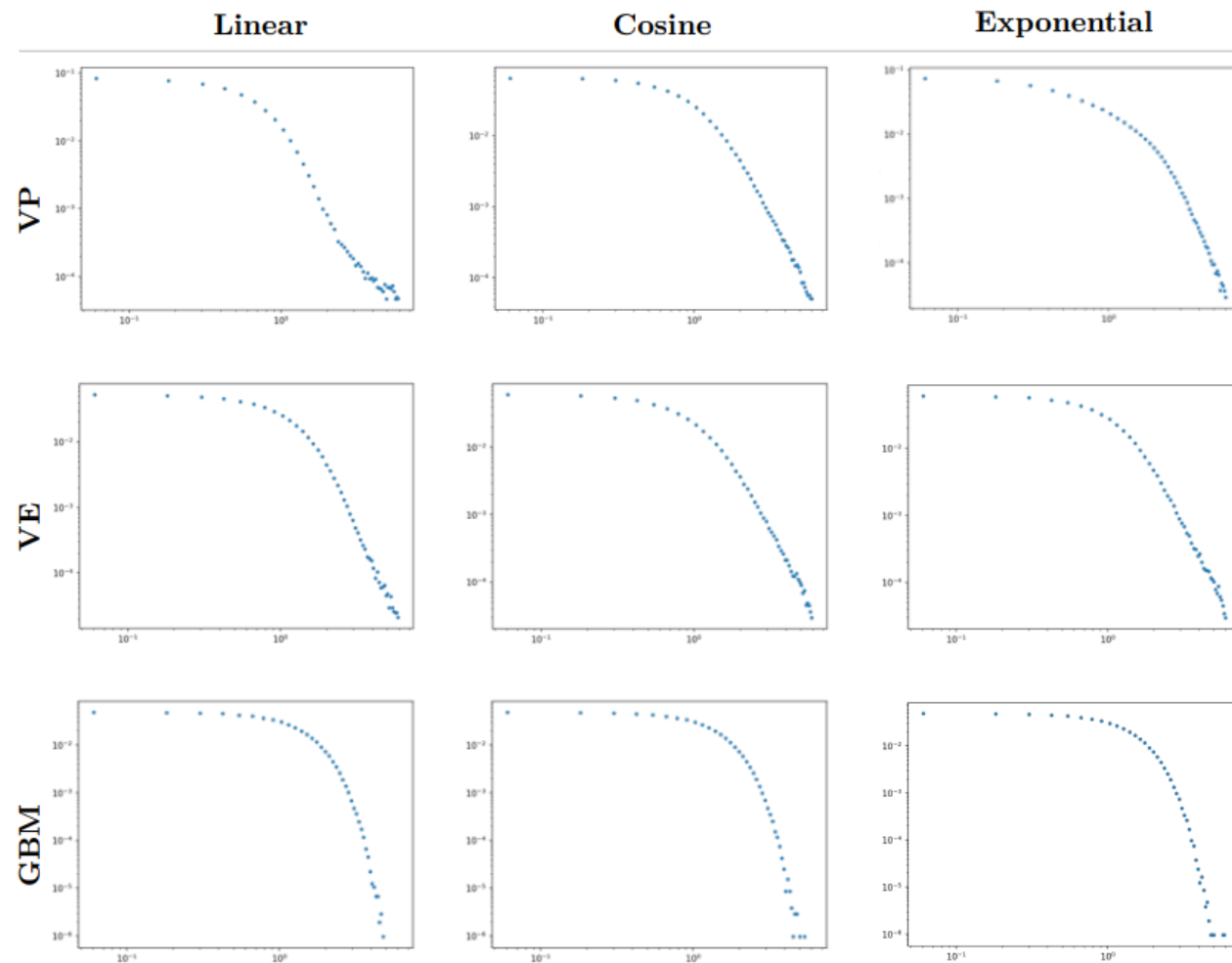


Figure B.5: Heavy-tail distribution (log-log scale, positive tail) for all nine Phase 1 configurations. *Rows* (top to bottom): VP, VE, and GBM SDE families. *Columns* (left to right): linear, cosine, and exponential noise schedules. The estimated power-law tail exponent $\hat{\alpha}$ is reported in Tab. 6.2; the empirical reference is $\hat{\alpha} = 4.378$.

Stylized Facts Phase 1: Volatility Clustering

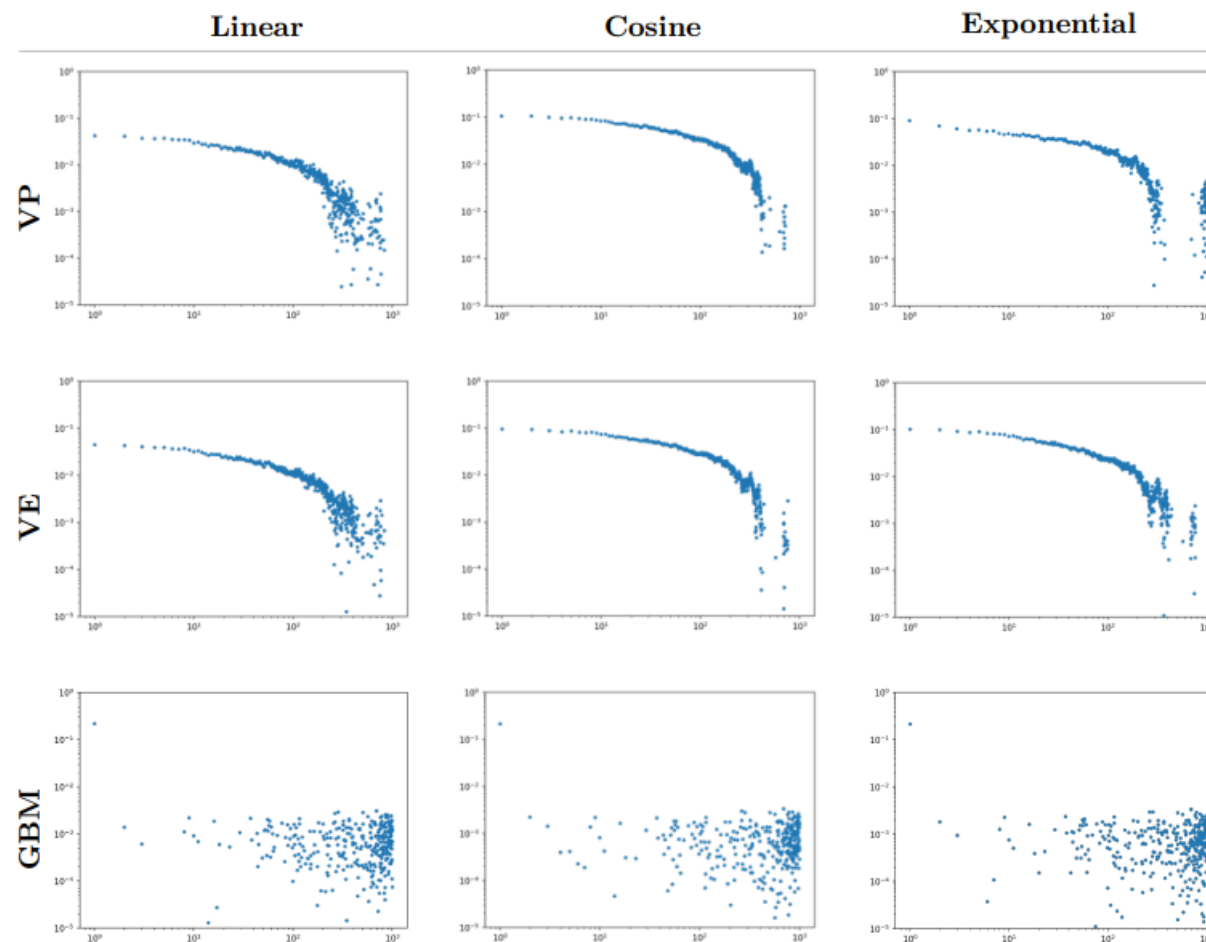


Figure B.6: Volatility clustering (autocorrelation of absolute returns on a log-log scale) for all nine Phase 1 configurations. Rows (top to bottom): VP, VE, and GBM SDE families. Columns (left to right): linear, cosine, and exponential noise schedules. Slow power-law decay indicates successful reproduction of long-range volatility dependence; rapid collapse or flattening indicates failure.

Stylized Facts Phase 1: Leverage Effect

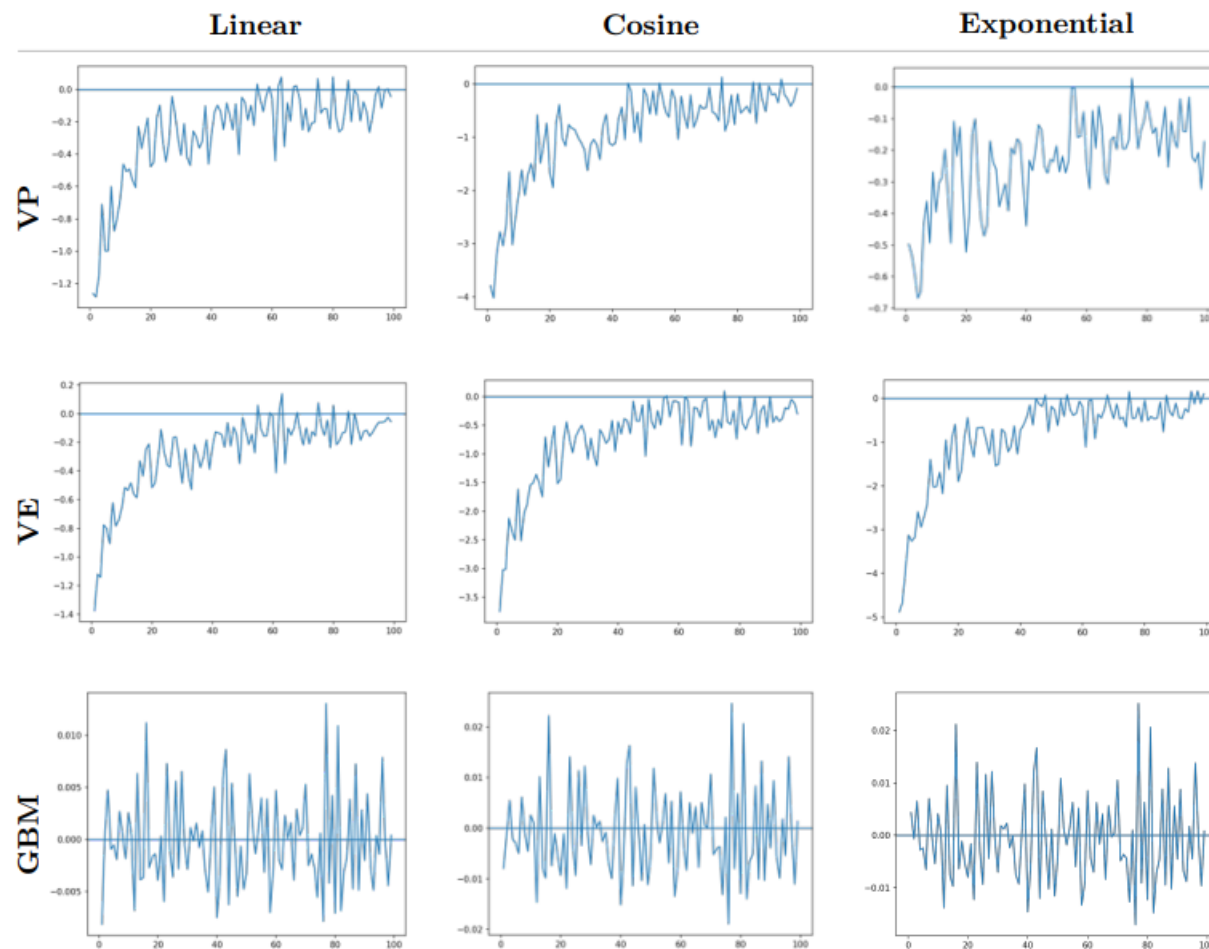


Figure B.7: Leverage effect (lead-lag correlation $L(k)$ between returns and future squared returns) for all nine Phase 1 configurations. Rows (top to bottom): VP, VE, and GBM SDE families. Columns (left to right): linear, cosine, and exponential noise schedules. Persistent negative values at small positive lags indicate successful reproduction of the leverage effect; oscillation around zero indicates failure.

Phase 2: EDM-CSDI CRPS

Configuration	Split	N. Samples	MAE ↓	CRPS ↓	T_1^{norm}	T_1	T_2^{norm}
Climatology	Val	25	0.619 13	0.448 69	0.897 45	0.018 73	0.897 53
OLS	Val	25	0.253 62	0.186 92	0.373 85	0.007 80	0.373 86
EDM-CSDI (Cond.)	Val	25	0.186 00	0.121 44	0.244 38	0.005 10	0.245 88
Climatology	Both	25	0.662 88	0.482 32	0.964 66	0.020 13	0.964 29
OLS	Both	25	0.285 20	0.212 52	0.425 07	0.008 87	0.425 10
EDM-CSDI (Cond.)	Both	25	0.195 46	0.126 02	0.257 31	0.005 37	0.262 58
Climatology	Val	50	0.608 51	0.449 00	0.897 82	0.018 73	0.897 65
OLS	Val	50	0.253 62	0.186 90	0.373 85	0.007 80	0.373 90
EDM-CSDI (Cond.)	Val	50	0.184 41	0.121 41	0.244 31	0.005 10	0.245 84
Climatology	Both	50	0.650 65	0.482 35	0.964 61	0.020 13	0.964 52
OLS	Both	50	0.285 20	0.212 55	0.425 15	0.008 87	0.425 20
EDM-CSDI (Cond.)	Both	50	0.193 79	0.126 59	0.257 27	0.005 37	0.262 59

Table 6.6: CRPS decomposition for the EDM-CSDI models against the climatology and OLS baselines. CRPS is decomposed as $\text{CRPS} = T_1^{\text{norm}} - \frac{1}{2}T_2^{\text{norm}}$, where T_1 is the accuracy term unnormalised by $\sigma = 0.020867$. **N. Samples** is to be intended for each window. Lower CRPS is better.

Phase 2: EDM-CSDI Coverage Interval

Split	Nominal (%)	EDM-CSDI 25	EDM-CSDI 50
Val	50.0	50.59	52.23
Val	80.0	78.41	82.16
Val	90.0	86.66	90.35
Val	95.0	91.34	94.06
Val	99.0	93.46	96.89
Both	50.0	51.57	53.05
Both	80.0	78.95	82.52
Both	90.0	86.87	90.50
Both	95.0	91.46	94.02
Both	99.0	93.60	96.84

Table 6.7: Empirical coverage at nominal levels for EDM-CSDI with 25 samples, and EDM-CSDI with 50 samples. Values closer to the nominal level indicate better-calibrated prediction intervals.

Phase 2: EDM-CSDI Price Paths comparison

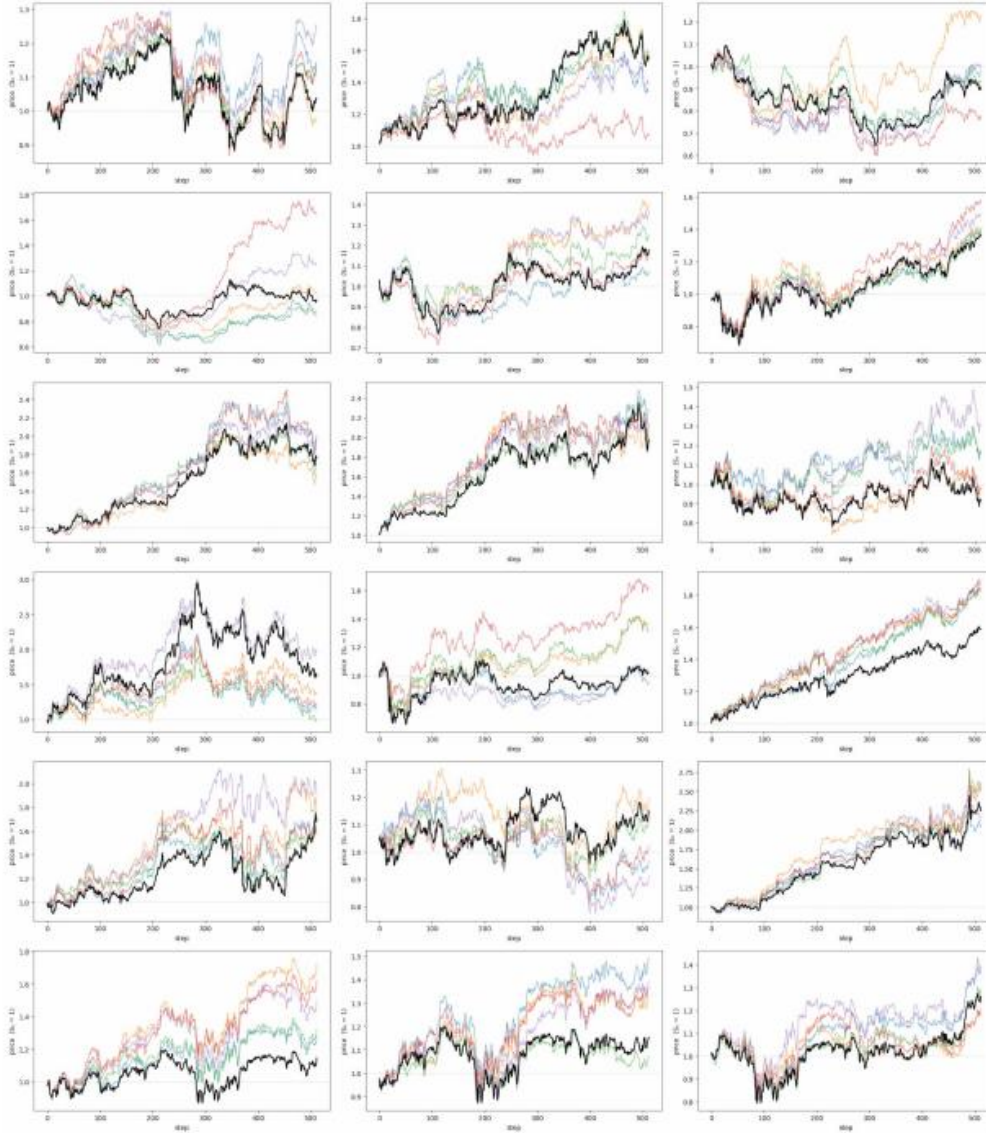


Figure B.8: Sample price paths for EDM-CSDI (Con.).

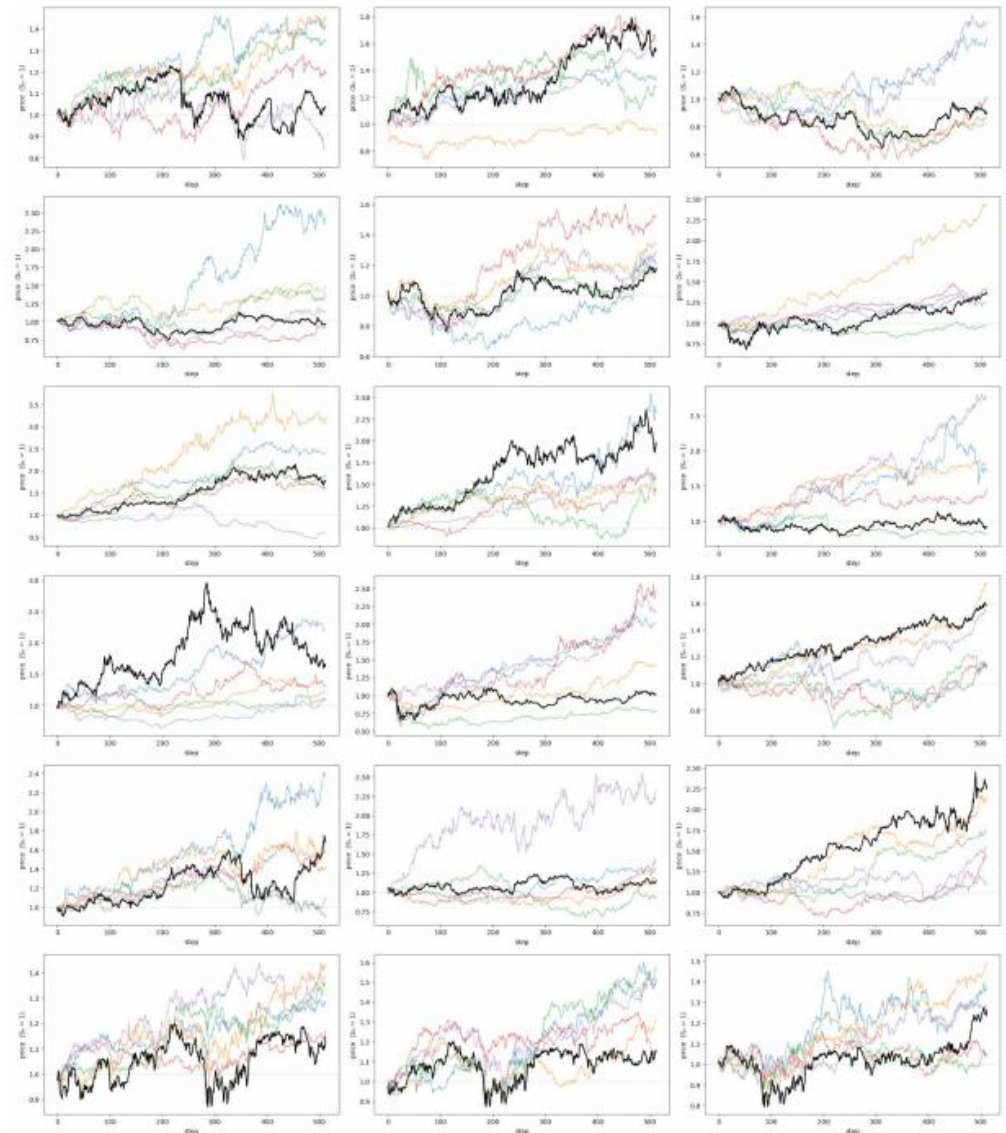


Figure B.9: Sample price paths for EDM-CSDI (Unc.).

Phase 2: EDM-CSDI Price Paths Aggregate

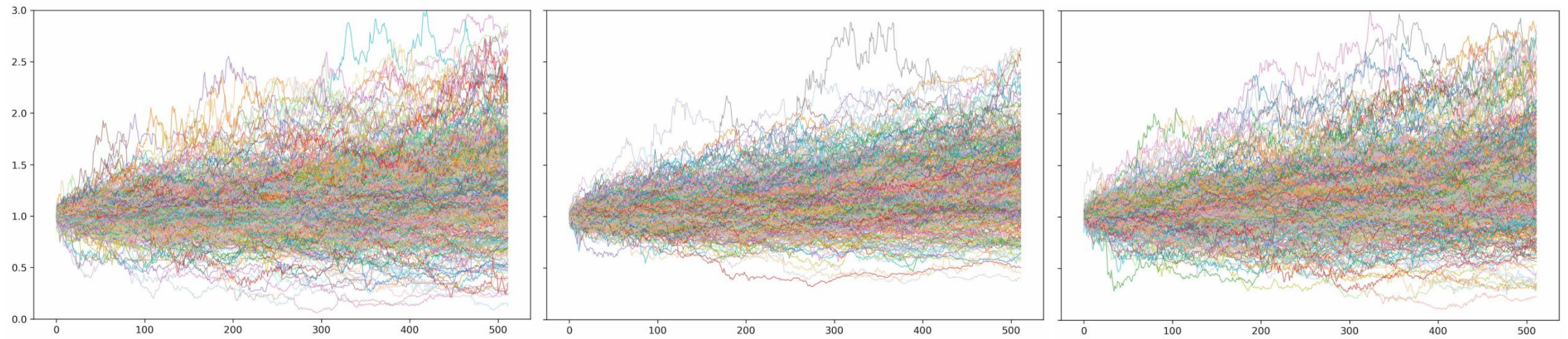


Figure 6.12 and Figure B.10: Left. Reference. Center. EDM-CSDI (Unc.). Right. EDM-CSDI (Cond.)

Volatility Analysis: Volatility Proxy Table

Mean volatility proxy	LOW	MEDIUM	HIGH
Reference	0.0138	0.0177	0.0235
EDM-CSDI (Unc.)	0.0120	0.0143	0.0173
EDM-CSDI (Cond.)	0.0131	0.0166	0.0232
Gen/Ref (Unc.)	0.87	0.81	0.74
Gen/Ref (Cond.)	0.95	0.94	0.99

Table C.7: Mean volatility proxy (flat 512-step std, unnormalised log-return units) by reference regime, and the corresponding generated/reference ratios for EDM-CSDI (Cond.) and (Unc.).